

A differentiable PyTorch implementation of the OpenIFS land-surface physics

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Abstract. Land-surface schemes are mature in process content but are written in Fortran, confining parameter and sensitivity studies to ensembles of independent integrations. We present a differentiable PyTorch implementation of the complete OpenIFS SURF land-surface physics that preserves the tiled process ordering of the Fortran reference and keeps every conditional regime switch unsmoothed, so the differentiated operator is the verified one. It matches the reference at every duration tested: a 30-day controlled integration of the 97,709-column CMFD domain ends at surface-temperature and snow-water-equivalent RMSE of 6.90×10^{-6} K and 1.05×10^{-6} kg m⁻², a one-year autonomous integration holds terminal relative errors below 5.1×10^{-3} %, and ten successively restarted years (2009–2018) show no error growth; CUDA-graph execution on an NVIDIA A100 sustains a 3.8-fold speed-up over the 32-thread Fortran reference. Against independent Terra MODIS observations, the configuration reproduces nighttime land surface temperature (bias 0.66 K) but carries a -6.4 K daytime cold bias. Decomposing this bias with reverse-mode derivatives verified against finite differences to 4×10^{-4} , we find that eight global parameter scales explain at most 0.5 K of it, that the remainder is a structural 8.1 K peak-to-peak deficit of near-surface diurnal amplitude, and that a two-timescale surface-layer closure designed from this diagnosis and verified in state space removes 67–84 % of the bias (calibration to held-out) in every month of the year and ports back to the Fortran driver without re-tuning. These results show how a differentiable land surface moves evaluation beyond numerical fidelity to observation-constrained process improvement.

15 *Copyright statement.* TEXT

1 Introduction

Land-surface schemes regulate the exchange of momentum, energy, water, and carbon between the atmosphere and the terrestrial surface. Their treatment of soil moisture, snow, vegetation, and surface temperature influences near-surface meteorology, boundary-layer development, runoff, and the memory of land–atmosphere anomalies.

20 Land-surface modelling rests on several decades of process development accumulated in a small number of community schemes. The Common Land Model, which assembled the land physics of earlier biosphere–atmosphere schemes into a single column model for energy, water, and carbon exchange (Dai et al., 2003), is widely used in climate and Earth system modelling

and has been carried forward along two main lines: the Community Land Model (CLM) developed in the United States (Oleson et al., 2010) and the Common Land Model (CoLM) developed in China, most recently documented as CoLM 2024 (Dai et al., 2004; Yuan and Dai, 2025). The Noah scheme and its multi-parameterization successor Noah-MP carry the land surface of mesoscale numerical weather prediction (Niu et al., 2011). Operational numerical-weather-prediction centres have likewise long developed their own land-surface models, among them ECMWF’s ECLand, the Joint UK Land Environment Simulator (JULES; Wiltshire et al., 2020), and the Community Atmosphere–Biosphere Land Exchange model (CABLE; Haverd et al., 2018). At ECMWF, in particular, the land surface evolved from the original TESSEL formulation through the tiled HTESSEL hydrology (Viterbo and Beljaars, 1995; van den Hurk et al., 2000; Balsamo et al., 2009) into ECLand, a modular modelling system that couples energy, water, and carbon and underpins the land surface of the operational IFS and its ensemble reanalyses (Boussetta et al., 2021). The OpenIFS land-surface component examined here belongs to this lineage; its physical formulation is documented in the IFS physical-process documentation (ECMWF, 2024). These schemes are mature in their process content, but they are written in Fortran, and neither derivatives nor accelerator-oriented batch execution is part of their design, so parameter and sensitivity studies rely on ensembles of independent integrations.

Atmospheric prediction has in the same years moved toward models whose complete time integration is differentiable. Differentiable programming was brought into Earth-system modelling to combine process models with learned components inside one gradient computation (Gelbrecht et al., 2023), and reviews of differentiable geoscience identify gradient access through the process equations as the requirement that unifies physical and machine-learning approaches (Shen et al., 2023). Neural-GCM couples a differentiable dynamical core with learned parameterizations and trains the combination end to end (Kochkov et al., 2024); its precipitation-focused successor is trained directly against satellite retrievals, which is only possible because every component of the model can be differentiated (Yuval et al., 2026). These developments show what differentiability offers operational prediction: the observation record itself becomes the training signal.

Differentiable land-surface implementations are more recent and remain site-focused. JAX-CanVeg reimplements the CanVeg canopy model in JAX, keeps the ecohydrological process chain intact, and embeds a neural-network stomatal-conductance closure within the process equations (Jiang et al., 2025). NoahPy recasts the Noah land surface model, in a permafrost thermohydrology configuration, as a process-encapsulated recurrent structure in PyTorch; it reproduces the Fortran original at site level with Nash–Sutcliffe efficiency above 0.99, and gradient-based calibration of its soil parameters converges faster and more stably than the SCE-UA search at a Qinghai–Tibet Plateau site (Tian et al., 2026). No differentiable implementation yet covers the complete land physics of an operational forecasting system over a continental domain with decadal verification, and the conditioning of the resulting estimation problem, which decides whether a gradient-based fit is well posed, has received little attention in land-surface applications.

Land-surface simulations now appear in applications that require large ensembles, parameter sensitivities, or the assimilation of land-surface observations. These applications benefit from an implementation that evaluates many independent columns concurrently and provides derivatives of selected model outputs with respect to inputs and parameters. Finite-difference ensembles need one integration per perturbed control; automatic differentiation obtains every selected derivative from a single reverse calculation while retaining process-based state variables and conservation constraints.

This study presents a PyTorch implementation of the OpenIFS SURF land-surface physics. The implementation retains the tiled process sequence of the reference scheme: surface exchange, the implicit surface energy balance, canopy interception, soil heat and water transport, and single-layer snow processes. Every conditional regime switch of the reference scheme is kept unsmoothed, so the operator that is differentiated is the same operator whose numerical equivalence is verified. Its purpose is to reproduce the numerical behaviour of SURF while enabling batched CPU/GPU execution and differentiation of the tensor operations. The implementation is therefore evaluated as a numerical translation of an established physical scheme rather than as a new land-surface parameterization. Verification and estimation are given equal weight: equivalence against the Fortran reference is established before any gradient is used.

The study has six objectives. First, the principal SURF equations and their tensor representation are described. Second, the implementation is evaluated against the independent Fortran reference implementation under identical roughness and surface-layer exchange boundary conditions, a separate native OpenIFS offline-column experiment isolates differences associated with reconstructing the atmospheric surface-layer boundary in standalone mode, and a one-year full-domain autonomous integration over heterogeneous CMFD forcing tests the complete prognostic sequence from a common initial state. Third, a ten-year successively restarted integration examines the stability of this equivalence over multiple forcing years. Fourth, multi-step automatic derivatives of forcing, initial state, and vegetation controls are verified against centred finite differences and applied in regional sensitivity analysis, observation-space identifiability diagnosis, parameter calibration, and initial-state retrieval. Fifth, the autonomous integration is evaluated against independent Terra MODIS observations of land surface temperature and snow cover. Sixth, and central to the scientific aim, the differentiability is used to close the loop between the MODIS observations and the process formulation: the residual daytime bias is separated into a parameter-explained part and a structural remainder, the missing process is diagnosed by reverse-mode attribution, and a new finite-capacity surface-layer closure is designed, calibrated, and verified against the same observations. The batch-size CPU and GPU experiments that quantify the acceleration of the batched tensor implementation close the study, without changing the physical time step or numerical precision.

Section 2 describes the model formulation. Section 3 presents the Fortran–PyTorch consistency experiments, including the ten-year successively restarted integration of Sect. 3.4. Section 4 describes the differentiable formulation and its use in sensitivity analysis and initial-state retrieval. Section 5 presents the independent MODIS evaluation and the structural daytime bias it reveals, then uses the differentiable formulation to attribute that bias to a missing near-surface process and to design, calibrate, and verify a finite-capacity surface-layer closure that removes most of it; the parameter-calibration and pseudo-forcing experiments that establish the structural nature of the bias and motivate the closure are documented in Appendix C. Section 6 reports computational scaling.

2 Model background and governing equations

OpenIFS SURF represents the land surface as a set of independent vertical columns coupled to the atmosphere through surface fluxes. Each column is partitioned into surface tiles, so the prognostic state is naturally written as tile variables and layer

90 variables,

$$\mathbf{x} = \{\mathbf{T}_{\text{skin}}, \mathbf{T}_{\text{soil}}, \mathbf{W}_{\text{soil}}, S_{\text{snow}}, f_{\text{snow}}, W_{\text{skin}}, T_{\text{snow}}, \rho_{\text{snow}}, \alpha_{\text{snow}}\}. \quad (1)$$

Here $\mathbf{T}_{\text{skin}}$ is the tile skin temperature, $\mathbf{T}_{\text{soil}}$ and $\mathbf{W}_{\text{soil}}$ are the four-layer soil temperature and water stores, S_{snow} is snow water equivalent, f_{snow} is the diagnosed snow-cover fraction, W_{skin} is intercepted canopy or skin water, and T_{snow} , ρ_{snow} , and α_{snow} are snow temperature, density, and albedo. Grid-box diagnostics are obtained by tile area weighting,

$$95 \quad \bar{q} = \sum_{i=1}^{N_{\text{tile}}} f_i q_i, \quad \sum_{i=1}^{N_{\text{tile}}} f_i = 1, \quad (2)$$

where f_i is the fractional area of tile i (Viterbo and Beljaars, 1995; Balsamo et al., 2009; Boussetta et al., 2021). This tiled structure is preserved explicitly, and the snow and wet-skin tile fractions are updated after each time step.

2.1 Atmospheric forcing and offline surface coupling

In coupled OpenIFS forecasts, SURF receives lower-atmosphere state variables, near-surface exchange coefficients, and vertical diffusion boundary quantities from the atmospheric physics. In offline mode these quantities must instead be constructed from meteorological forcing. The forcing vector used by the offline PyTorch implementation is

$$\mathbf{y} = \{T_a, q_a, p_s, u_a, v_a, R_{\text{SW}}^{\downarrow}, R_{\text{LW}}^{\downarrow}, P_{\text{rain}}, P_{\text{snow}}\}, \quad (3)$$

with time interpolation or accumulation conventions matching the OpenIFS offline configuration. The surface stress and scalar increments are then closed by the roughness-length update and the surface-layer vertical-diffusion closure of the IFS physics package before the land update is invoked. This closure is part of the offline PyTorch model only. When SURF is coupled back into a complete atmospheric model, these quantities are supplied by the atmospheric vertical-diffusion scheme rather than being recomputed within SURF (ECMWF, 2024).

2.2 Surface energy balance

For each active tile, the skin temperature is advanced from a surface energy balance,

$$110 \quad C_i \frac{dT_{s,i}}{dt} = R_{n,i} - H_i - L_v E_i - G_i - M_i, \quad (4)$$

where C_i is the effective surface heat capacity, $R_{n,i}$ is net radiation, H_i is sensible heat flux, $L_v E_i$ is latent heat flux, G_i is the conductive heat flux into the soil or snow pack, and M_i is the latent heat sink or source associated with melt and freezing. The net radiation is

$$R_{n,i} = (1 - \alpha_i) R_{\text{SW}}^{\downarrow} + \epsilon_i R_{\text{LW}}^{\downarrow} - \epsilon_i \sigma T_{\text{rad},i}^4, \quad (5)$$

115 where α_i is tile albedo, ϵ_i is emissivity, σ is the Stefan–Boltzmann constant, and $T_{\text{rad},i}$ is the radiative skin temperature used by the atmospheric model. The original scheme solves the skin temperature implicitly over the coupling time step, which is retained in the PyTorch implementation (Viterbo and Beljaars, 1995; ECMWF, 2024).

2.3 Turbulent exchange

The turbulent fluxes use a resistance or bulk-transfer form (Louis, 1979; ECMWF, 2024),

$$120 \quad H_i = \rho_a c_p C_{H,i} |\mathbf{V}| (T_{s,i} - T_a), \quad E_i = \rho_a C_{E,i} |\mathbf{V}| (q_{s,i} - q_a), \quad (6)$$

where ρ_a is air density, c_p is the heat capacity of air, $C_{H,i}$ and $C_{E,i}$ are stability- and roughness-dependent exchange coefficients, $|\mathbf{V}|$ is the near-surface wind speed, and $q_{s,i}$ and q_a are surface and atmospheric specific humidity. Momentum exchange is represented analogously through the drag coefficient C_D . In the tensor formulation these algebraic coefficient evaluations are expressed as element-wise differentiable maps, while the discrete regime switches for snow, bare soil, vegetation, and wet-skin

125 tiles are retained unchanged.

2.4 Canopy interception and wet skin

The intercepted water reservoir evolves according to

$$\frac{dW_{\text{skin}}}{dt} = P_{\text{int}} - E_{\text{int}} - D_{\text{int}}, \quad (7)$$

where P_{int} is intercepted precipitation, E_{int} is evaporation from the wet canopy or skin reservoir, and D_{int} is drainage or excess
130 water released once the tile-dependent capacity is exceeded. The reservoir is bounded by a vegetation-dependent maximum storage $W_{\text{skin}}^{\text{max}}(LAI, f_{\text{veg}})$, and the wet-skin fraction is diagnosed from $W_{\text{skin}}/W_{\text{skin}}^{\text{max}}$. This diagnostic affects the tile fractions and therefore feeds back onto the turbulent and radiative flux weighting (Viterbo and Beljaars, 1995; van den Hurk et al., 2000).

2.5 Soil heat

The soil thermal state follows a four-layer diffusion equation (Viterbo and Beljaars, 1995; ECMWF, 2024),

$$135 \quad C_k \frac{\partial T_k}{\partial t} = \frac{\partial}{\partial z} \left(\lambda_k \frac{\partial T}{\partial z} \right) + S_k, \quad k = 1, \dots, 4, \quad (8)$$

where C_k is volumetric heat capacity, λ_k is effective thermal conductivity, and S_k contains surface and phase-change source terms. The equation is discretized in finite-volume form with inter-layer conductive fluxes $\lambda_{k+1/2}(T_{k+1} - T_k)/(z_{k+1} - z_k)$; the implicit tridiagonal solve and the layer geometry are preserved from the reference implementation (Viterbo and Beljaars, 1995; ECMWF, 2024).

140 2.6 Soil hydrology

HTESSEL updates soil water through infiltration, evapotranspiration, percolation, and runoff (Balsamo et al., 2009). For layer k ,

$$\Delta z_k \frac{\partial \theta_k}{\partial t} = I_{k-1/2} - I_{k+1/2} - E_k - R_k, \quad (9)$$

where θ_k is volumetric soil water, I is vertical liquid-water flux, E_k is root-zone water extraction, and R_k is surface or sub-
 145 surface runoff. The prognostic water mass $W_k = \rho_w \Delta z_k \theta_k$ is bounded by soil-type-dependent residual, wilting-point, field-
 capacity, and saturation values. These bounds keep the update numerically stable and preserve the hydrological constraints of
 the reference model.

2.7 Snow mass, snow temperature, density, and cover

The snow pack is represented as a single-layer store with prognostic mass, temperature, density, and albedo (Viterbo and
 150 Beljaars, 1995; Dutra et al., 2010; ECMWF, 2024). Snow water equivalent obeys

$$\frac{dS_{\text{snow}}}{dt} = P_{\text{snow}} - E_{\text{snow}} - M_{\text{snow}}, \quad (10)$$

where P_{snow} is snowfall, E_{snow} is sublimation, and M_{snow} is melt. Snow temperature is advanced by the same surface energy
 balance as Eq. (4), written on the snow tile with the snow-pack heat capacity C_{snow} and the melt energy sink $L_f M_{\text{snow}}$. Snow
 density evolves through compaction and fresh-snow updates, while snow albedo is aged and refreshed by snowfall. The active
 155 OpenIFS configuration in this study uses the density-scaled snow-cover diagnostic,

$$f_{\text{snow}} = \min(1, 10 S_{\text{snow}} / \rho_{\text{snow}}), \quad (11)$$

with very small snow fractions clipped to zero for tile-partition stability. This relation is central to the interpretation of regional
 maps because full snow cover can occur with moderate SWE when the density-scaled threshold is exceeded (ECMWF, 2024;
 Boussetta et al., 2021).

160 2.8 Differentiable tensor formulation

The full time step is expressed as a single nonlinear operator

$$\mathbf{x}^{n+1} = \mathcal{M}_{\Delta t}(\mathbf{x}^n, \mathbf{y}^{n:n+1}, \mathbf{s}; \boldsymbol{\theta}), \quad (12)$$

where $\mathbf{s}$ contains static land descriptors and $\boldsymbol{\theta}$ contains physical parameters or lookup-table values. Most of the constituent
 updates are element-wise differentiable maps. The scheme, however, is not a smooth dynamical system: rain–snow partitioning,
 165 tile activation, soil-moisture bounds, wet-skin capacity overflow, freezing thresholds, and snow-cover clipping are conditional
 statements that partition the state space into regimes with distinct governing expressions.

The translation treats this structure as follows: every branch is kept exactly as in the reference, and none is smoothed,
 relaxed, or replaced by a differentiable surrogate. The resulting operator is differentiable almost everywhere, and reverse-
 mode differentiation through it returns the exact derivative of the realized computation on the active branch; at a regime
 170 crossing the derivative is the one-sided derivative of that branch, a local linearization of the switched dynamics. This choice
 has a methodological consequence: the differentiated operator is the same operator whose numerical equivalence is verified
 in Sect. 3, so any gradient-based inference refers to the discrete IFS physics itself rather than to a smoothed reconstruction of
 it. Smoothing the switches would yield better-behaved gradients but would break the equivalence with the reference solution.

The verification of Sect. 4 shows that the retained branches do not degrade the derivatives: multi-step reverse-mode derivatives agree with centred finite differences to within 4×10^{-4} relative error, and the fraction of columns whose local response is dominated by a regime transition is mapped and reported (0.19–4.17 % depending on season) rather than omitted.

3 Fortran–PyTorch equivalence verification

The verification distinguishes the translated SURF process sequence from the standalone reconstruction of the atmospheric surface-layer boundary. In a coupled OpenIFS forecast, the surface roughness lengths and the surface-layer turbulent exchange coefficients are supplied by the atmospheric physics. In an offline integration they must instead be diagnosed from the meteorological forcing. The first experiment therefore holds all Fortran roughness, surface-layer exchange, and radiation boundary arrays fixed when integrating the SURF update with PyTorch under those boundaries. The second experiment lets the Fortran and PyTorch implementations construct their own roughness and surface-layer exchange states from the same forcing. This separation tests the numerical translation and the standalone interface without conflating the two questions.

All regional experiments use the 97,709 valid land columns of the 2018 China Meteorological Forcing Dataset (CMFD) at its native 0.1° grid (He et al., 2020). The common restart, static fields, and decoded three-hourly forcing records are byte-identical between the two paths. The physics time step is 1800 s and all prognostic calculations use double precision. The controlled experiment records the Fortran-generated roughness, surface-layer exchange, and radiation inputs at every step; the PyTorch integration consumes those records without constructing an independent atmospheric closure. The initial states, observation subsets, reference outputs, and archived evaluation products of every experiment reported below are collected in the accompanying data package (Yang et al., 2026).

3.0.1 Common initial state.

Unless stated otherwise, every experiment in this paper, the controlled and autonomous equivalence experiments of this section, the multi-year integration, the MODIS evaluation, and the calibration, attribution, and closure experiments, starts from a single, shared land initial state. That state is produced by a five-year spin-up: the land-surface state of ERA5-Land on 1 January 2005 (four-layer soil temperature and volumetric soil water, snow water equivalent, snow temperature, density, albedo, and skin temperature) is mapped onto the 97,709-column domain, and the offline model is integrated continuously over 2005–2009 under CMFD forcing so that the state equilibrates with that forcing; the state reached at the end of 2009 is the common initial condition.

3.1 Controlled full-domain boundary-driven integration

The 30-day controlled boundary-driven integration is a direct end-to-end test of the translated SURF sequence: tiled exchange, radiative and surface-energy calculations, canopy and wet-skin stores, four-layer soil heat and water updates, and single-layer snow processes receive the same state and boundary arrays as the Fortran run. Table 1 reports every saved prognostic quantity at day 30. Relative RMSE is normalized by the root-mean-square Fortran value over the stated comparison mask.

Table 1. Day-30 controlled full-domain Fortran–PyTorch comparison with identical captured roughness, surface-layer exchange, and radiation boundaries. “All” denotes all 97,709 valid land columns. The $f_{\text{snow}} > 10^{-3}$ mask is used for snow temperature, density, and albedo because those diagnostic variables are ill-conditioned as snow mass and heat capacity vanish.

Quantity (unit)	mask	RMSE	relative RMSE (%)
Surface temperature (K)	all	6.90×10^{-6}	2.68×10^{-6}
Soil temperature, layer 1 (K)	all	4.04×10^{-3}	1.55×10^{-3}
Soil temperature, layer 2 (K)	all	9.95×10^{-4}	3.79×10^{-4}
Soil temperature, layer 3 (K)	all	1.49×10^{-4}	5.65×10^{-5}
Soil temperature, layer 4 (K)	all	3.11×10^{-5}	1.18×10^{-5}
Soil water, layer 1 (kg m^{-2})	all	6.09×10^{-4}	2.34×10^{-3}
Soil water, layer 2 (kg m^{-2})	all	4.87×10^{-4}	6.47×10^{-4}
Soil water, layer 3 (kg m^{-2})	all	2.41×10^{-4}	1.28×10^{-4}
Soil water, layer 4 (kg m^{-2})	all	8.81×10^{-6}	2.92×10^{-6}
Snow water equivalent (kg m^{-2})	all	1.05×10^{-6}	1.20×10^{-5}
Snow temperature (K)	$f_{\text{snow}} > 10^{-3}$	2.40×10^{-4}	9.54×10^{-5}
Snow density (kg m^{-3})	$f_{\text{snow}} > 10^{-3}$	1.85×10^{-4}	1.30×10^{-4}
Snow albedo (-)	$f_{\text{snow}} > 10^{-3}$	1.63×10^{-8}	2.67×10^{-6}

205 The resolved-snow criterion retains 69,005 columns at day 30. Without that criterion, the all-column snow-temperature RMSE rises to 0.00321 K because the largest contributions occur at nearly snow-free columns, where the PyTorch and Fortran SWE values are both below $2.5 \times 10^{-5} \text{ kg m}^{-2}$ and both snow fractions are below 1.6×10^{-6} . It is therefore a diagnostic-temperature amplification in a vanishing snow store rather than an energetically relevant snow-cover difference. The unfiltered statistic is retained in the archived metrics, while the resolved-snow statistic is used where snow temperature, density, and
210 albedo are interpreted physically.

Figure 1 shows the continuous error trajectory. The curves are not separate daily forecasts: both models retain the same prognostic state through the full 30 days. The domain errors remain near output precision through approximately day 21, after which a few threshold-sensitive columns contribute visible excursions. The day-30 maps in Fig. 2 show that the typical spatial differences remain small compared with the state amplitudes.

215 The 30-day and one-year experiments serve different verification purposes. The former is a boundary-controlled test of the translated sequence: identical Fortran-generated roughness, surface-layer exchange, and radiative boundary fields remove atmospheric-closure differences and allow the translated SURF process sequence to be examined directly. The latter is a system-level offline test: each implementation constructs its own boundary state from the common forcing and is therefore sensitive to both the SURF translation and the standalone atmospheric-surface closure. The two experiments are consequently
220 complementary rather than repeated integrations.

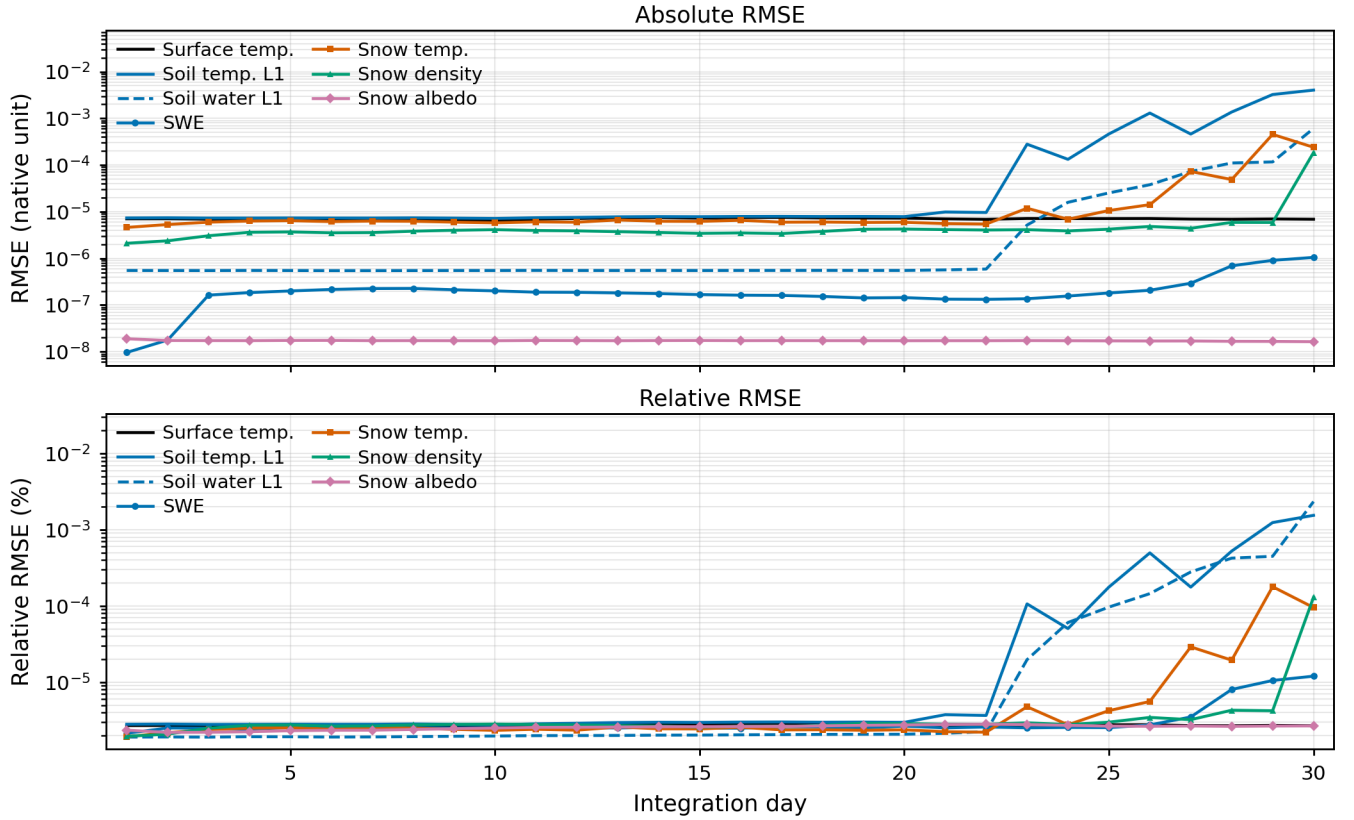


Figure 1. Error evolution during the continuous 30-day controlled full-domain boundary-driven integration. Every PyTorch step uses Fortran-captured roughness, surface-layer exchange, and radiation boundaries. Absolute and relative RMSE use the masks described in Table 1.

3.2 One-year autonomous offline integration

The controlled experiment verifies the SURF process sequence given the atmospheric boundary. The more demanding autonomous experiment verifies the complete offline pathway in which each implementation constructs its own roughness and surface-layer exchange state. Both models start from the same common spun-up initial state described above and use the same 2018 CMFD forcing, 1800 s time step, three-hour forcing interpolation, and precipitation/radiation alignment. The 97,709-column integration runs for 17,520 continuous steps (365 d), with states archived every 480 steps (10 d) and at the terminal step. No non-finite values occur in any saved PyTorch prognostic field.

Fortran writes state snapshots every 480 steps through day 360, and a separate Fortran integration with the same initial state, forcing, and numerical settings writes the day-365 terminal state, differing only in output frequency. PyTorch stores the same 10-day sequence plus the exact terminal step. Thus Fig. 3 describes one continuous year-long integration sampled for diagnosis, not a set of 10-day forecasts.

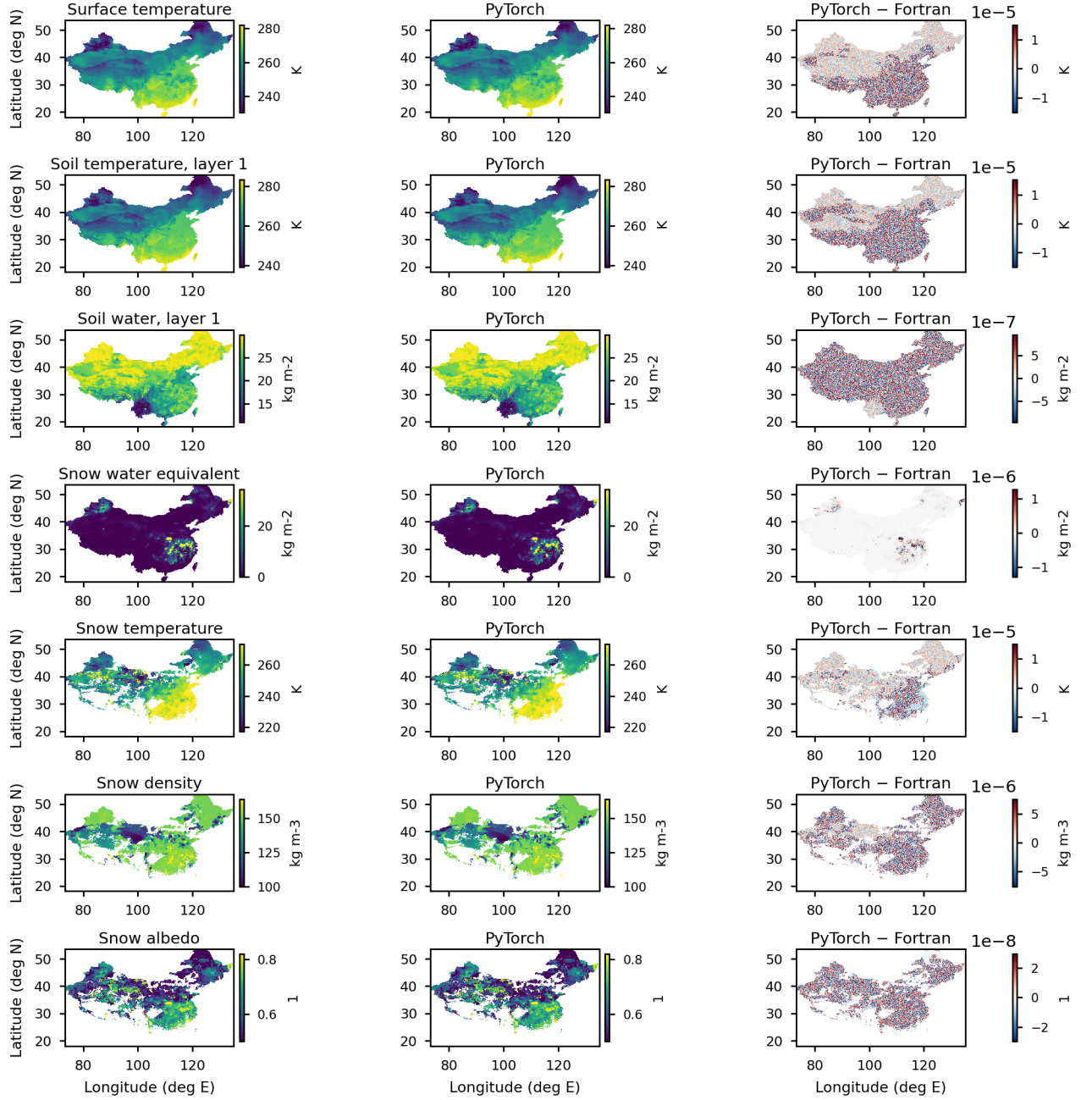


Figure 2. Day-30 controlled full-domain comparison on the original CMFD grid. Each row is one field; columns show the Fortran state, PyTorch state, and signed PyTorch minus Fortran difference. Snow diagnostic maps use the resolved-snow mask. Error colour limits are robust percentile limits.

Table 2. Day-365 Fortran–PyTorch comparison for the autonomous offline CMFD integration. Relative RMSE is normalized by the root-mean-square Fortran field over the stated mask. “All” contains 97,709 columns; the resolved-snow mask contains 82,112 columns at day 365.

Quantity (unit)	mask	RMSE	relative RMSE (%)
Soil temperature, layer 1 (K)	all	2.83×10^{-4}	1.07×10^{-4}
Soil temperature, layer 2 (K)	all	2.38×10^{-4}	8.94×10^{-5}
Soil temperature, layer 3 (K)	all	1.72×10^{-4}	6.32×10^{-5}
Soil temperature, layer 4 (K)	all	1.22×10^{-4}	4.34×10^{-5}
Soil water, layer 1 (kg m^{-2})	all	4.19×10^{-4}	2.43×10^{-3}
Soil water, layer 2 (kg m^{-2})	all	1.36×10^{-3}	2.54×10^{-3}
Soil water, layer 3 (kg m^{-2})	all	6.22×10^{-3}	3.24×10^{-3}
Soil water, layer 4 (kg m^{-2})	all	5.55×10^{-3}	1.03×10^{-3}
Snow water equivalent (kg m^{-2})	all	7.15×10^{-4}	2.61×10^{-4}
Snow temperature (K)	$f_{\text{snow}} > 10^{-3}$	2.30×10^{-4}	9.08×10^{-5}
Snow density (kg m^{-3})	$f_{\text{snow}} > 10^{-3}$	7.22×10^{-3}	5.02×10^{-3}
Snow albedo (-)	$f_{\text{snow}} > 10^{-3}$	3.72×10^{-6}	6.44×10^{-4}

As a reference-side reproducibility check, the archived one-year Fortran experiment was repeated on the same machine from the independently regenerated forcing file and an identical configuration: all 24 output fields of the repeat integration, including every soil, snow, and tile state, are bitwise identical to the archive. The Fortran reference trajectory used in the comparisons
235 below is therefore deterministic and exactly reproducible, so any Fortran–PyTorch difference is attributable to the translation and not to run-to-run variability of the reference.

The PyTorch side is checked with the same rigour. Repeated integrations from a given restart are bitwise identical, and re-integrating the first 30 d of the multi-year integration of Sect. 3.4 through the same code path as the experiments reproduces the archived records bit for bit on both accelerator architectures used in this study (NVIDIA RTX A6000 and A100).
240 Reproducibility does, however, extend only as far as the pinned software stack. A decade integration launched under a different PyTorch build (2.6.0+cu124 instead of the pinned 2.11.0+cu130; all arithmetic remains float64) departs from the pinned-build trajectory at last-bit rounding level within the first days (day-10 snow-water maximum difference $7.2 \times 10^{-8} \text{ kg m}^{-2}$). The offline snow–soil feedback then amplifies these last-bit seeds to about $1.1 \times 10^{-3} \text{ kg m}^{-2}$ rms snow water and locally up to 2.8 K skin temperature by the end of the first year, with unchanged physics and identical inputs. Every experiment reported in this
245 paper therefore runs in a single pinned environment (Python 3.10, PyTorch 2.11.0+cu130), and the manifest of each archived experiment records the interpreter, the PyTorch and CUDA builds, and SHA-256 hashes of every source file, in addition to the input and output digests.

The annual error evolution is shown in Fig. 3. Differences remain small throughout the cold-season spin-up and increase locally during the spring snow evolution, when round-off-scale differences can traverse different density, phase, or cover
250 thresholds. They do not produce a monotonic domain-wide drift: at day 365 the largest relative RMSE among the prognostic

fields is $3.2 \times 10^{-3}\%$ (layer-3 soil water), the resolved-snow diagnostics reach $5.0 \times 10^{-3}\%$ (snow density), and all soil-temperature relative RMSE values are below $1.1 \times 10^{-4}\%$.

Table 2 reports the pinned source revision whose SHA-256 hashes are recorded in the experiment manifest; this is the same revision as the decadal integration of Sect. 3.4. An earlier code state of the same implementation, since superseded and
 255 not byte-archived, reached about ten times smaller terminal RMSE (3.5×10^{-4} K layer-1 soil temperature). Comparing the two trajectories column by column localizes the effect of the code change: at day 10, 93% of the 97,709 columns are bitwise identical (median absolute difference 6×10^{-14} K) and only 59 columns differ by more than 10^{-3} K (maximum 3.3×10^{-2} K), all of them at states that sit on discrete phase, snow-cover, or stability thresholds. The residual Fortran–PyTorch difference of the autonomous configuration is therefore governed by round-off-seeded selection among discrete physical branches, not
 260 by formula or unit discrepancies, and its amplitude depends on the exact source revision; the controlled boundary-driven integration of Sect. 3.1, which prescribes the atmospheric boundary, is insensitive to this mechanism.

3.3 Annual water and energy-budget diagnostics

The absence of non-finite values is supplemented by an explicit column water budget. For each 1800 s step, the diagnostic accumulates total precipitation, the signed surface moisture flux, surface and subsurface runoff, and changes in the four-layer
 265 soil, snow, and skin-reservoir water stores. The discrete residual is

$$\epsilon_W = \Delta S - (P + E - R_s - R_d), \quad (13)$$

where evaporation is negative when it removes water from the column. Over the full domain and year, the mean terms are $P = 634.78$, $E = -511.05$, $R_s = 64.16$, $R_d = 8.96$, and $\Delta S = 50.62$ kg m⁻². The terminal residual has an RMS of 5.90×10^{-12} kg m⁻² and a maximum absolute value of 3.45×10^{-11} kg m⁻², corresponding to $3.76 \times 10^{-13}\%$ of the mean annual
 270 water-throughput scale. This diagnostic is calculated from the prognostic update and is not fed back into the integration.

The lower-left panel of Fig. 7 reports the associated surface-energy diagnostic. The two summary measures are

$$\begin{aligned} \overline{\epsilon_E} &= -0.123 \text{ W m}^{-2}, \\ |\overline{\epsilon_E}| &= 3.87 \text{ W m}^{-2}, \end{aligned} \quad (14)$$

where the first quantity is the signed annual domain mean and the second is the mean absolute instantaneous residual. The latter is retained as a diagnostic measure because the implicit surface-energy solve, tile shortwave transmission, longwave
 275 reweighting, and snow-under-vegetation correction are evaluated in separate discrete stages. It is therefore not interpreted as an additional prognostic source or sink.

3.4 Decadal autonomous integration

The one-year autonomous comparison is extended to ten continuous CMFD years (2009–2018, 366-day years included), each year restarted from the terminal state of the previous year with the forcing linked continuously across the year boundaries. No
 280 error growth with lead time occurs. At every annual terminal state, soil-temperature relative RMSE stays below $8 \times 10^{-4} \%$

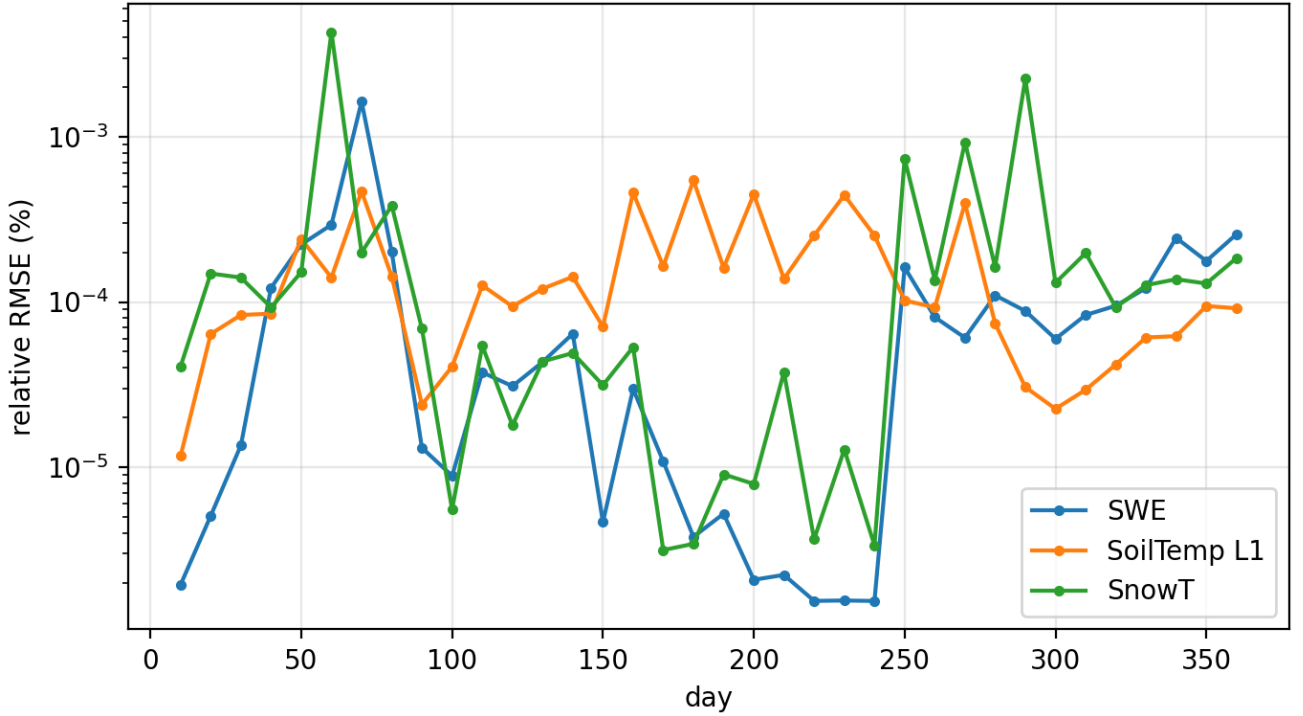


Figure 3. Absolute and relative Fortran–PyTorch error evolution during the continuous one-year autonomous offline integration. The saved 10-day states and day-365 terminal state are samples from the same integration. Three representative fields are shown: snow water equivalent, layer-1 soil temperature, and snow temperature. The fields not plotted stay within the following ranges over the year: layers 2–4 of soil temperature, 6.4×10^{-6} – 4.8×10^{-4} %; layers 1 and 4 of soil water, 2.4×10^{-6} – 1.3×10^{-2} %; snow density and albedo on the resolved-snow mask, 7.7×10^{-6} – 1.9×10^{-2} % and 2.3×10^{-6} – 6.4×10^{-3} % respectively. Snow temperature is masked at a snow-cover fraction above 10^{-3} .

across the decade, soil-water relative RMSE below 2.5×10^{-2} %, and snow-water relative RMSE below 3×10^{-3} %, the same magnitude as the one-year values of Table 2. The oscillation is the signature of a forced dissipative system with threshold-branch amplification: a round-off difference can flip a discrete branch decision and produce a finite jump, which soil diffusion and the surface-energy feedback then damp, rather than chaotic error growth, which is absent by construction because the atmospheric forcing is prescribed and identical in the two implementations; each field’s error peaks in its own threshold-active season, and the terminal-error level is a quasi-stationary balance between weather-set branch injection and diffusive damping, orders of magnitude below the seasonal variability of the fields themselves. The terminal-state comparison table, the annual error evolution, and the forcing alignment that motivates the exact restart procedure are collected in Appendix A.

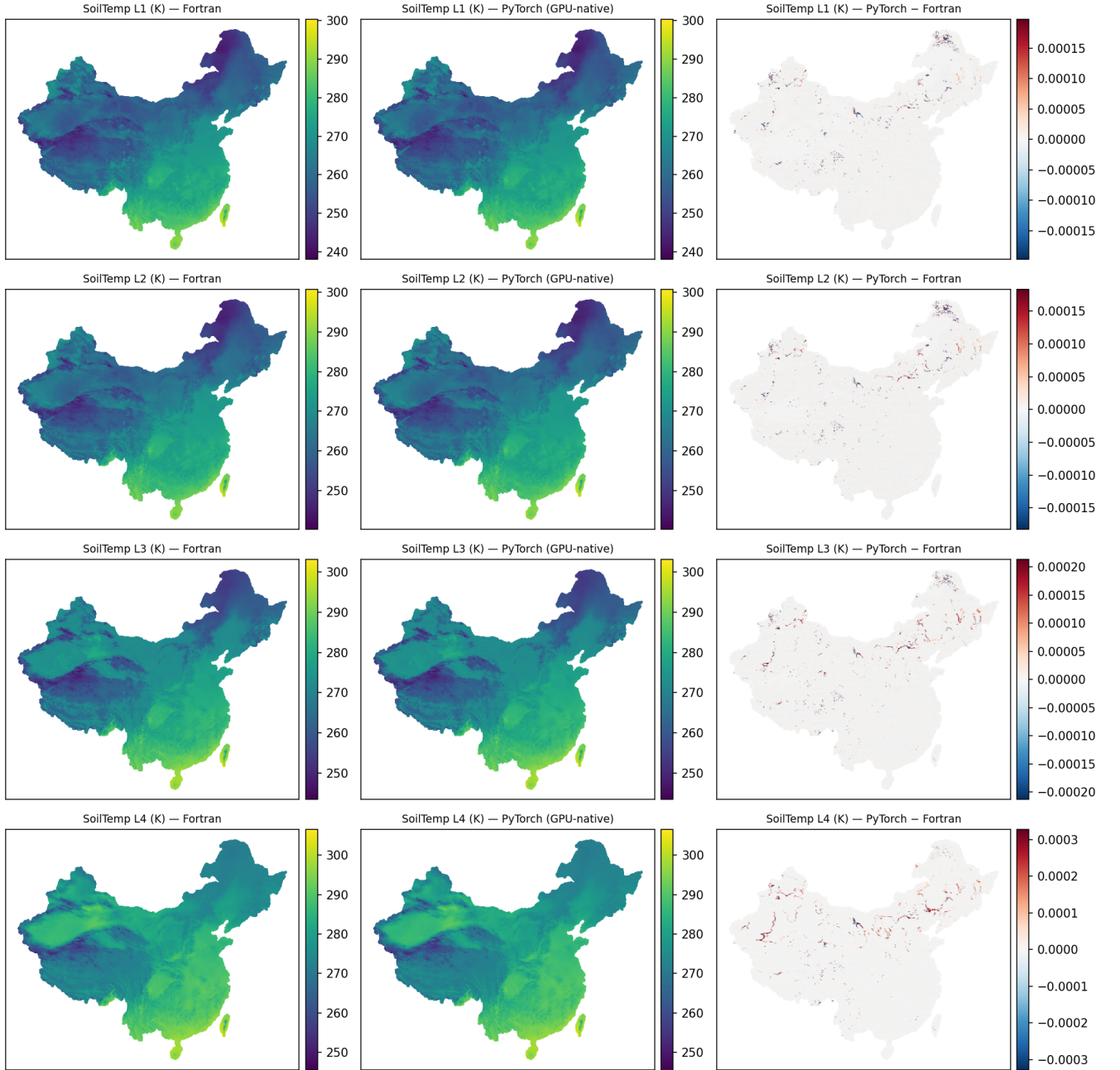


Figure 4. Day-365 full-domain comparison for all four soil-temperature layers. Each row is one layer; columns show the Fortran state, PyTorch state, and signed PyTorch minus Fortran difference. Difference colour limits use the 99.5th percentile of the absolute error.

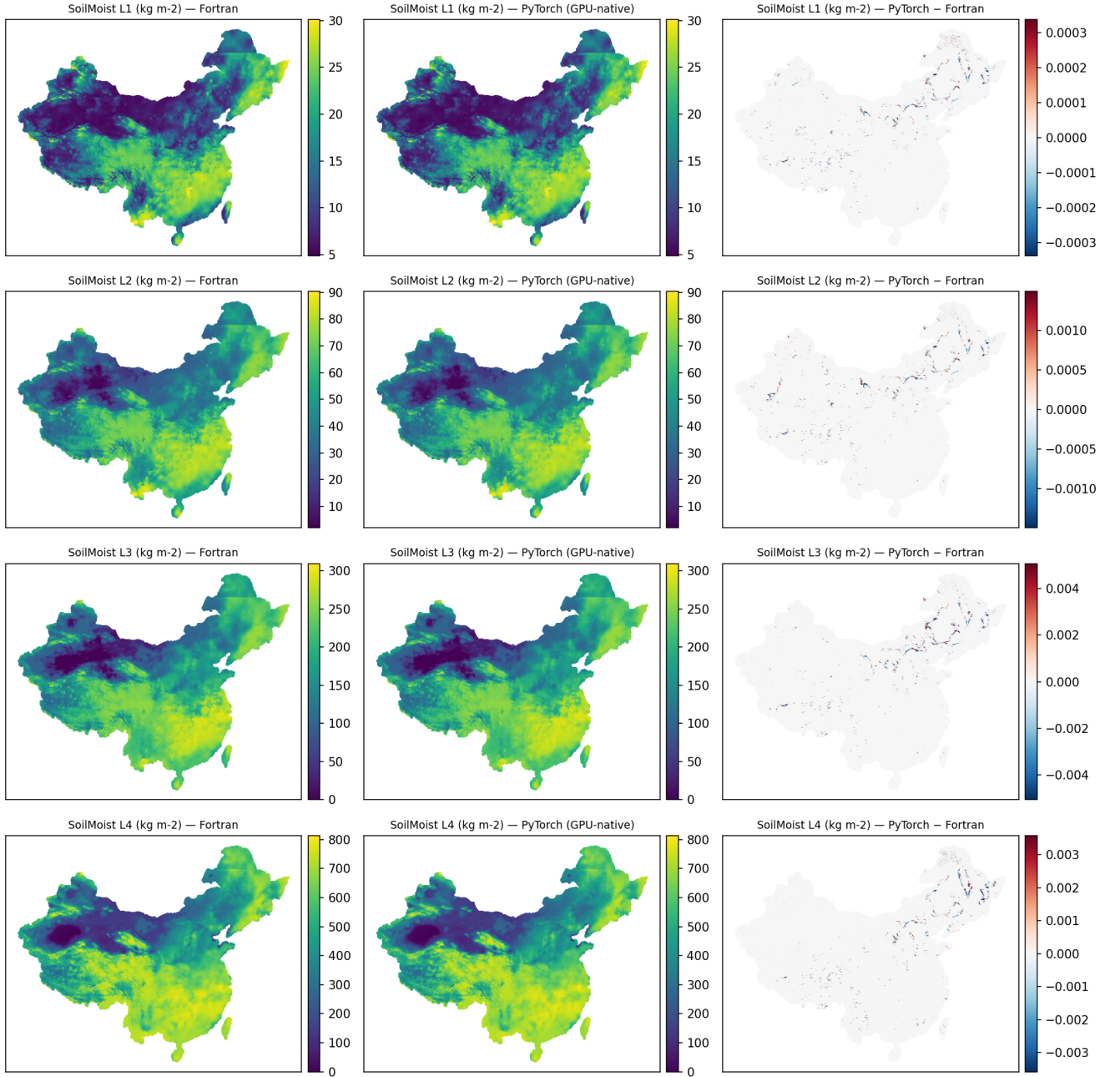


Figure 5. Day-365 full-domain comparison for all four soil-water layers. Each row is one layer; columns show the Fortran state, PyTorch state, and signed PyTorch minus Fortran difference. Difference colour limits use the 99.5th percentile of the absolute error.

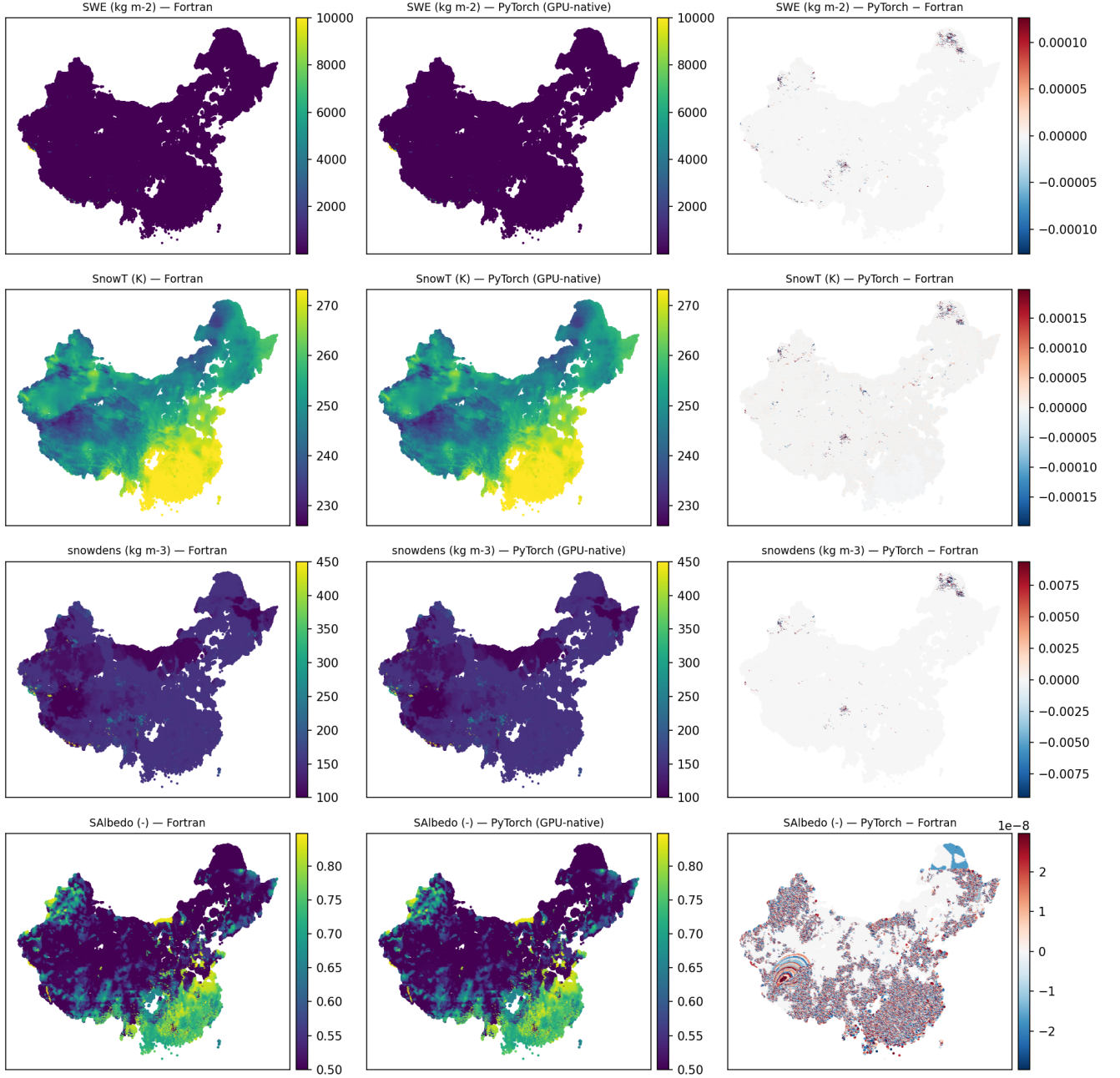


Figure 6. Day-365 full-domain comparison for snow water equivalent, snow temperature, snow density, and snow albedo. Each row is one field; columns show the Fortran state, PyTorch state, and signed PyTorch minus Fortran difference. The snow diagnostic fields use the $f_{\text{snow}} > 10^{-3}$ mask. Difference colour limits are robust percentile limits.

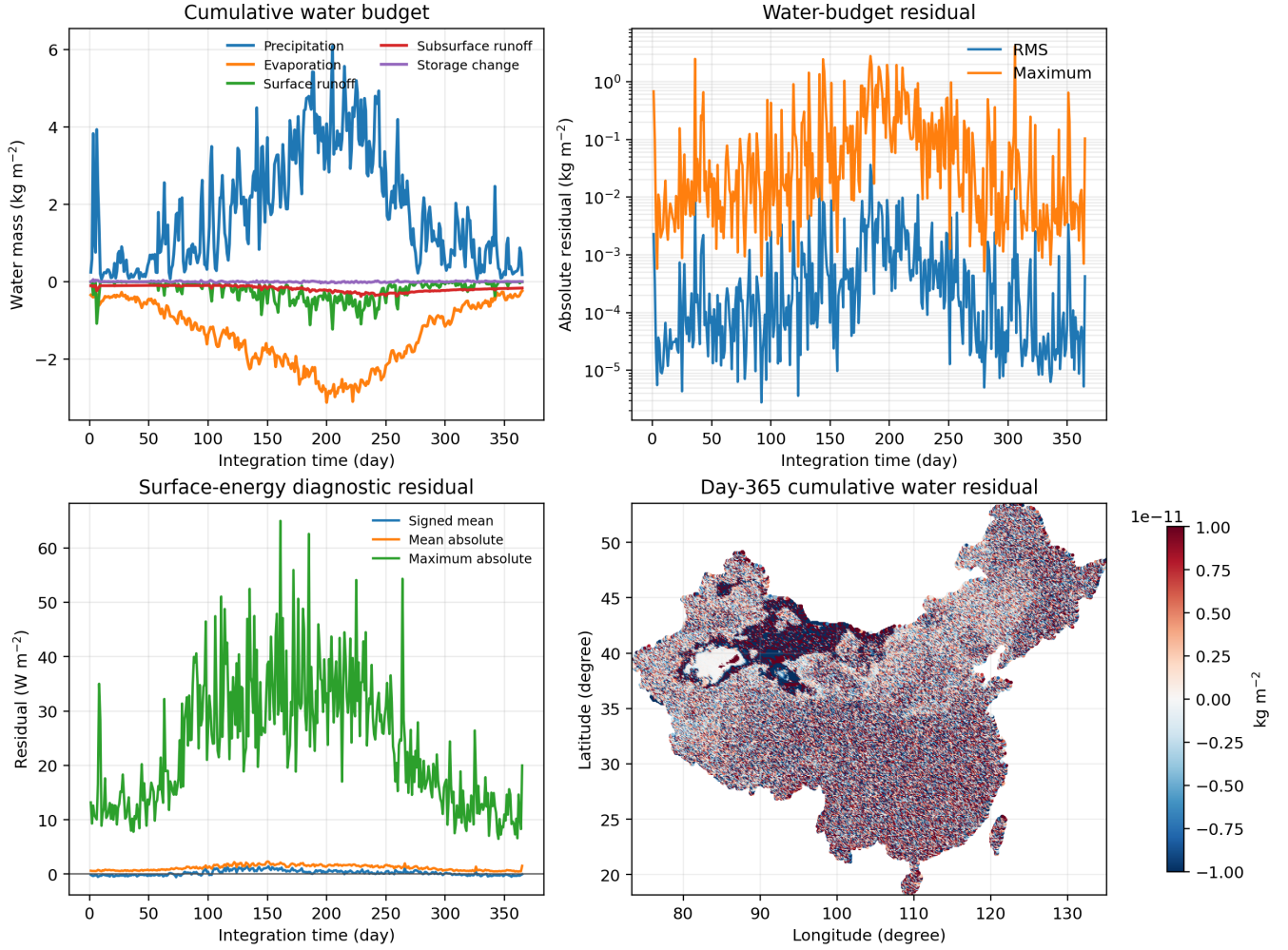


Figure 7. Annual water and energy-budget diagnostics for the continuous 97,709-column Torch integration. Upper left: domain-mean cumulative water terms, with outward fluxes plotted negative. Upper right: cumulative water residual. Lower left: signed and absolute surface-energy diagnostic residual. Lower right: day-365 cumulative water residual on the original CMFD longitude–latitude locations.

Table 3. Verification of 24 h multi-step automatic-differentiation derivatives against centred finite differences. The reported finite-difference step is the value with the smallest relative discrepancy in the tested range.

Control and loss	automatic derivative	relative error	ϵ
$\Delta T_{\text{air}} \rightarrow \overline{T}_{\text{skin}}^{24h}$	9.23691×10^{-1}	1.33×10^{-5}	10^{-2} K
$\Delta W_{\text{soil},1}^0 \rightarrow \overline{W}_{\text{soil},1}^{24h}$	7.27416×10^{-1}	9.05×10^{-5}	3×10^{-4}
$\text{LAI} \rightarrow E_{24h}$	-3.51626×10^{-1}	3.66×10^{-4}	10^{-4}

4 Differentiability and sensitivity analysis

290 The tensor formulation permits reverse-mode automatic differentiation through an unrolled SURF trajectory: for a scalar loss $\mathcal{J}$, derivatives can be evaluated with respect to forcing components, initial states, and selected parameters. For a loss depending on a time window, the backward pass returns all control-variable derivatives at the cost of one reverse calculation, rather than one additional forward integration per control, which is what makes regional, multi-parameter process diagnosis and constrained parameter estimation tractable. This section verifies the derivative (Sect. 4.1) and uses it for response diagnostics
 295 (Sect. 4.2); an initial-state retrieval against an independently executed Fortran trajectory is provided in Appendix B, and the observation-space application, identifiability, calibration, and structural attribution against MODIS, follows in Sect. 5.

4.1 Multi-step gradient verification

Automatic differentiation must be verified independently before its gradients are used to infer physical controls. A deterministic 24 h idealized summer diurnal experiment with four identical land columns, hourly coupling, no precipitation, and the complete
 300 autonomous SURF process sequence provides three multi-step controls: a uniform air-temperature forcing offset (loss: terminal mean skin temperature), a top-layer soil-water initial-state offset (loss: terminal top-layer soil water), and the active vegetation-class LAI table value (loss: 24 h accumulated evaporation). For each control, the reverse-mode derivative was compared with a centred finite difference over five perturbation magnitudes from 10^{-4} to 10^{-2} .

Table 3 shows agreement better than 4×10^{-4} in relative error for all three derivatives; the residual difference reflects
 305 the finite, fixed number of nonlinear surface-layer and implicit-solver iterations in the discrete model, not a loss of gradient connectivity (the perturbation-size dependence is shown in Appendix B). The test also identified an inactive snow-throughfall division: tensor conditionals evaluate both operands, so a snow-free and rain-free column can carry a masked 0/0 division into the backward calculation despite a finite forward state. The inactive denominator is replaced by one while the active path retains the original expression, which leaves the forward solution unchanged and prevents a nonphysical NaN derivative.

310 4.2 Regional and seasonal response diagnostics

With no horizontal exchange in the standalone configuration, every column can carry an independent infinitesimal air-temperature offset δT_i , and the local terminal response $S_i = \partial T_{\text{skin},i} / \partial \delta T_i$ over a 6 h window of the 2018 CMFD forcing is obtained for all 97,709 columns from one forward trajectory and one reverse pass; a centred-difference map would require one perturbation

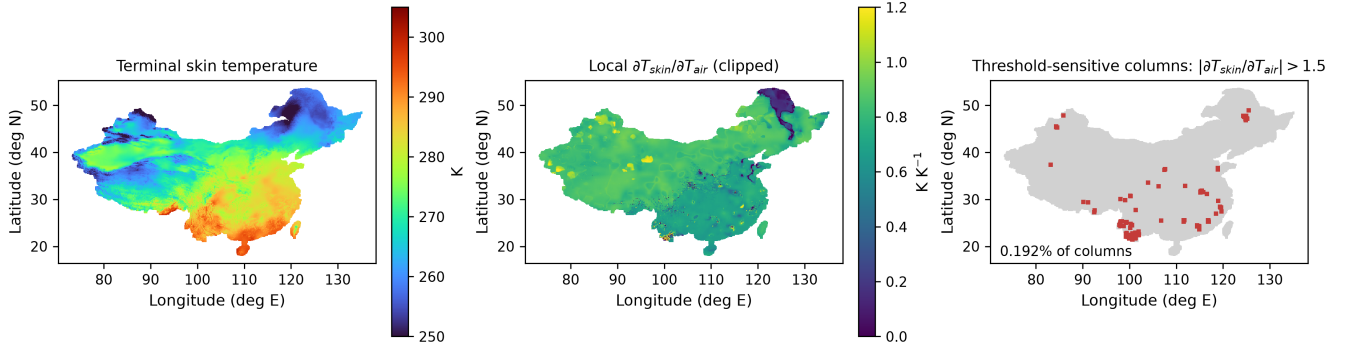


Figure 8. Six-hour regional automatic-differentiation diagnostic under the first 2018 CMFD forcing window from the common restart. Left: terminal grid-mean skin temperature. Centre: local response of terminal skin temperature to an independent near-surface air-temperature offset at each column; the scale is clipped only for visibility. Right: columns whose un-clipped absolute response exceeds 1.5 K K^{-1} .

Table 4. Seasonal 6 h local air-temperature response of terminal skin temperature over 97,709 CMFD land columns. The response is $S_i = \partial T_{\text{skin},i} / \partial T_{\text{air},i}$. Threshold-sensitive columns satisfy $|S_i| > 1.5 \text{ K K}^{-1}$.

Sample	median (K K^{-1})	5th–95th percentile (K K^{-1})	threshold-sensitive (%)
Late March	0.794	0.517–0.913	1.43
Late June	0.740	0.388–0.957	4.17
Late September	0.742	0.612–0.896	1.12
Late December	0.843	0.683–0.976	0.19

integration per column. Figure 8 shows the spatial heterogeneity: the median response is 0.818 K K^{-1} (5th–95th percentiles $0.557\text{--}0.907 \text{ K K}^{-1}$), so most columns retain a locally smooth, sub-unity skin-temperature response. A small set of columns (0.192 % in this window) exceeds 1.5 K K^{-1} in magnitude; these sit near threshold-sensitive surface-layer, saturation, or snow-regime states, are displayed explicitly rather than hidden by the clipped colour scale, and mark where a derivative is a local linearization rather than a finite-amplitude sensitivity.

Repeating the calculation at day-90, 180, 270, and 360 states of the annual integration, after a 24 h unperturbed re-equilibration of the fast variables, resolves the seasonal modulation (Table 4). The median response ranges from 0.740 K K^{-1} in late June to 0.843 K K^{-1} in late December; summer has both the broadest distribution and the largest fraction of threshold-sensitive columns (4.17 %), winter the most coherent response and the smallest fraction (0.19 %). The seasonal maps are provided in Appendix B. This separation between a typical smooth process response and a small threshold-dominated set is a property of the original parameterization, inherited rather than introduced by the translation, and is carried into the interpretation of every gradient result below.

4.3 Interpretation and use of gradients

The differentiable calculation retains the discrete physical regimes of the reference scheme rather than smoothing them, following the formulation of Sect. 2.8. Rain–snow partitioning, tile activation, saturation bounds, and snow-cover clipping are conditional statements, so the scheme is piecewise differentiable rather than smooth. A derivative is therefore local to the active regime, is a one-sided derivative at a regime crossing, and can be zero when a variable is limited by a bound or does not enter the selected process pathway, behaviour that reappears as the structurally zero and collinear control directions of Appendix C and is reported wherever gradient results are interpreted. Potential calibration parameters include soil hydraulic properties, snow albedo and density parameters, roughness-length scalings, and canopy storage parameters; the verified multi-step derivative developed here is the numerical foundation of the observation-space workflow of Sect. 5.

The Fortran comparisons establish numerical consistency, but agreement with a reference implementation does not establish physical realism. We therefore evaluated the autonomous 2018 Torch integration against independent Terra MODIS observations. Daily daytime and nighttime land surface temperature (LST) came from the 0.05° MOD11C1 collection 6.1 product (Wan et al., 2021). The collection-6 LST refinements are evaluated by Wan (2014). Fractional snow cover was taken from
340 MOD10C1 collection 6.1 (Hall and Riggs, 2021), whose data-record context is described by Riggs et al. (2017). Each of the 97,709 CMFD land-column centres was assigned the containing MODIS CMG cell. This samples the observations at the model locations but does not imply that the satellite and model spatial supports are identical.

The evaluation integration starts from the common spun-up land state described in Sect. 3 (ERA5-Land 1 January 2005 state spun up for five years to the end of 2009 under CMFD forcing). This spun-up restart is the initial state of every experiment in
345 this section and of the calibration, attribution, and closure experiments of Appendix C and Appendix D.

MOD11C1 reports the UTC view time separately for each cell. The native grid-mean radiative skin temperature of the model was sampled at the nearest completed 1800 s physics step. The maximum timing displacement is therefore 15 min. The primary, strict LST mask requires successful production, zero values in the mandatory and data-quality QC bits, and a stated LST error not exceeding 1 K. A broader sensitivity mask retains produced retrievals with a stated error not exceeding 2 K, includes many
350 more retrievals under the product’s “other quality” category, and is reported separately. MOD10C1 snow cover was compared with the prognostic SURF snow fraction at the nominal Terra 10:30 local-solar-time overpass, requiring a spatial quality flag of 0 or 1 and at least 50 % clear observations within the CMG cell. All regional moments and categorical scores are weighted by $\cos(\phi)$, and the observation operators are diagnostic and do not feed back into the integration.

Table 5. Area-weighted 2018 SURF evaluation against MODIS from the spun-up initial state. Bias is SURF minus MODIS. The July–December snow row gives the corresponding warm-season subset of the full-year snow statistics.

Quantity	mask or period	samples	model mean	observed mean	bias / RMSE / r
Daytime LST (K)	strict	3,727,377	294.87	301.22	−6.37 / 9.42 / 0.817
Nighttime LST (K)	strict	1,238,893	287.48	286.86	0.66 / 3.93 / 0.887
Daytime LST (K)	broader QC	19,857,571	286.51	289.89	−3.43 / 7.72 / 0.899
Nighttime LST (K)	broader QC	20,904,460	274.02	272.16	1.79 / 5.53 / 0.933
Snow fraction (-)	full year	16,997,616	0.0492	0.0764	−0.0217 / 0.192 / 0.575
Snow fraction (-)	July–December	8,642,926	0.0444	0.0624	−0.0177 / 0.173 / 0.566

The strict-QC results in Table 5 distinguish the daytime and nighttime behaviour. Nighttime LST has a small annual mean
355 bias of 0.66 K, an RMSE of 3.93 K, and a spatial-temporal correlation of 0.887. Daytime LST instead has a −6.37 K bias and a 9.42 K RMSE. The broader QC sample gives the same sign and seasonal structure but a smaller daytime bias of −3.43 K; the strict clear-sky sample is not a random subset of the land and atmosphere states, so both masks are shown rather than treated as an uncertainty range.

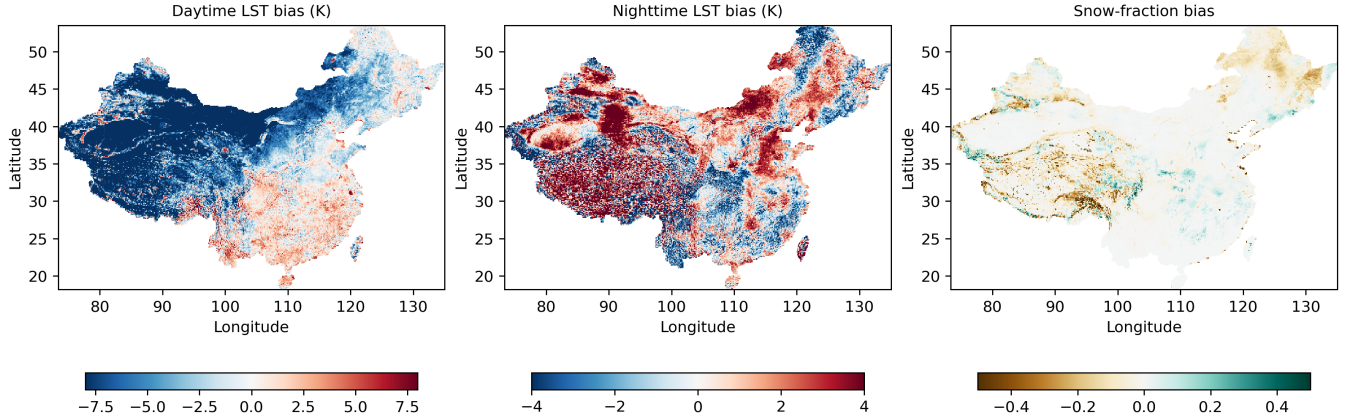


Figure 9. Annual mean SURF-minus-MODIS differences from the spun-up integration on the original CMFD longitude–latitude locations. LST maps use the strict QC mask; snow uses spatial QA 0–1 and at least 50 % clear coverage. Display limits are fixed at ± 8 K for daytime and ± 4 K for nighttime LST, and ± 0.5 for snow fraction; the nighttime range is half the daytime range, in proportion to the smaller bulk spread of the nighttime bias (interquartile range -1.3 to $+2.2$ K versus -8.3 to $+0.1$ K daytime). Untruncated aggregate statistics are reported in Table 5.

The error is localized by the surface–air contrast. Under the strict mask, the mean daytime surface-minus-air contrast is 4.88 K in SURF and 11.25 K in MODIS; at night the corresponding values are -2.95 and -3.61 K. Over snow-free columns (3.04×10^6 strict samples) the daytime bias grows monotonically with the observed contrast, from $+4.5$ K in the lowest decile to -17.7 K in the highest, and regionally from $+0.7$ K over the humid southeast to -8.3 K over the arid northwest and -8.0 K over the Tibetan Plateau. The modelled day–night skin-temperature range at the view times is 13.7 K against an observed 21.4 K. Where MODIS retrieves under the strict mask the CMFD forcing is itself predominantly clear (view-time transmissivity above 0.54 for 90 % of samples), so cloud-sampling inconsistency affects a small minority. The daytime cold bias is therefore a structural under-amplification of the daytime surface–air decoupling under strong insolation, concentrated where the observed contrast is largest; it is not a uniform offset, a masking artefact, or a night-time coupling error. It is also shared by the Fortran reference rather than introduced by the translation: the two implementations’ 3-hourly grid-mean skin-temperature archives agree over all 2,920 records of the year to a median absolute difference of 1.0×10^{-5} K and a 99th percentile of 4×10^{-3} K.

The model reproduces the broad seasonal retreat and re-establishment of snow cover but underestimates the observed area-weighted snow fraction: for the full year, fractional RMSE is 0.192 and correlation 0.575; at a 0.5 snow-cover threshold, the area-weighted probability of detection, false-alarm ratio, and critical success index are 0.316, 0.381, and 0.265. Figure 10 condenses the annual behaviour into monthly, area-weighted curves: the forecast follows the observed night-pass cycle closely, under-amplifies the warm-season day-pass maximum, and tracks the winter build-up and spring retreat of snow cover at a lower fraction than observed. Snow-cover extent is thus a further target for the observation-constrained experiments below; it is not evidence that the numerical translation itself is inaccurate.

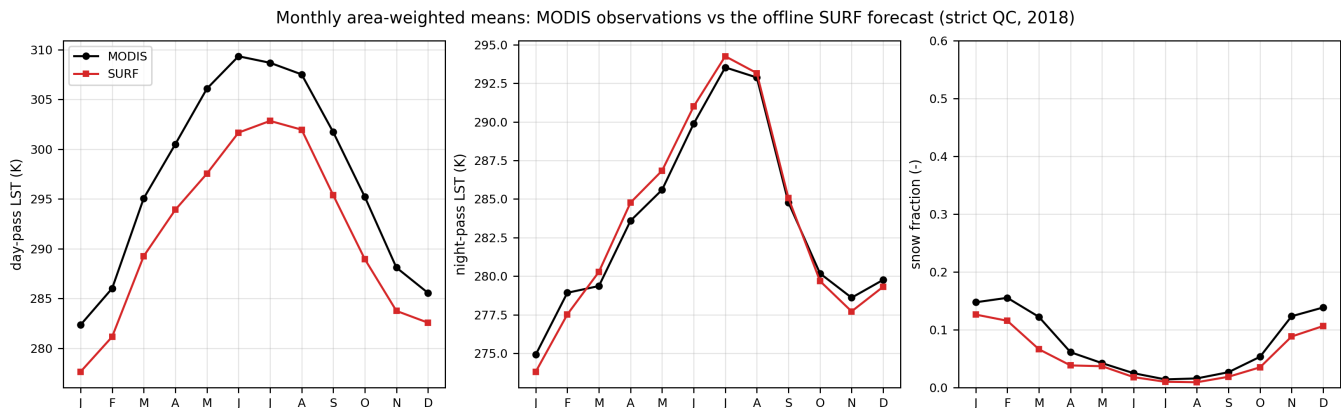


Figure 10. Monthly, area-weighted means of the MODIS observations (black) and the offline SURF forecast (red): day-pass LST, night-pass LST, and snow fraction (2018, spun-up initial state of Sect. 5; strict QC for LST, snow QA for snow, applied to both curves). The forecast follows the observed night-pass cycle to within 1.4 K, under-amplifies the warm-season day-pass maximum (forecast peak 6.5 K below observed), and reproduces the phase of the snow cycle at a lower fraction (peak 0.11 against 0.16).

5.1 Attribution of the daytime cold bias

Two checks localize the error before anything is corrected. First, global multiplicative parameter scales contribute only a limited correction: a gradient-based calibration of eight such scales (the vegetation leaf-area-index, stomatal-resistance, and roughness-length tables split by vegetation class, plus soil thermal-conductivity and dry-soil heat-capacity scales), trained on ten autumn–winter windows and evaluated on ten held-out spring–summer windows, removes only about 0.5 K of the 5.5–8.3 K windowed daytime offset (7–10 %). This is a statement about these eight global scales and about land surface temperature alone: it does not exclude a larger role for other parameters (hydrological, albedo, snow) or for other state variables. In particular, the tested control space contains no spatially heterogeneous static fields, such as soil texture and hydraulic properties or vegetation and albedo maps, and the ten-day anchored windows deliberately suppress the slow soil-water memory by restoring archived initial states. Such factors carry real uncertainty and may contribute materially to the daytime bias; the 0.5 K figure is therefore a bound on this specific control space, not a general verdict on parameter optimization, and an observing-system simulation with the identical operator, overpass times, and optimizer recovers imposed truth parameters to within 0.3–1.1 %, so the limited correction is not an inversion failure (Appendix C). Second, the dominant daytime error is associated with the offline atmosphere–surface exchange rather than with these parameter scales: when additive air-temperature offsets and multiplicative wind and shortwave scales are optimized instead of model parameters, the calibration-window day bias is largely removed, and the optimal profile, afternoon +4.8 K, evening −3.1 K, states what is missing: the offline exchange delivers about 8.1 K peak-to-peak less near-surface diurnal amplitude than the observed surface demands, or an equivalent radiative gain. The same corrections retain most of their effect in the held-out season under the anchored-day evaluation, so the profile is seasonally

robust; it remains a diagnosis rather than a correction, as a ten-control additive adjustment of one year's forcing has no process form (Appendix C).

The attribution experiment is enabled by the differentiable formulation. With the role of the model parameters bounded, the residual bias is interrogated from the forcing side: a pseudo-forcing correction, additive air-temperature offsets in the eight
400 local-time 3-hour bins plus multiplicative scales on wind and shortwave, ten controls in all, is optimized against the same MODIS loss. A finite-difference probe identifies the day-overpass bin and the shortwave scale as the dominant levers, and gradients of the loss with respect to all ten controls are obtained from a single reverse pass through the unrolled trajectory, so the full diurnal profile is adjusted jointly. Trained on the ten autumn–winter days, the optimization removes most of the calibration-window daytime bias (-5.48 to -1.44 K; RMSE 7.45 to 5.11 K) and converges to a definite profile, afternoon
405 $+4.8$ K, evening -3.1 K (Fig. 11). The profile constitutes the functional form of the missing process: the offline exchange under-delivers near-surface diurnal amplitude in the afternoon and over-delivers it in the evening, a deficit of about 8.1 K peak-to-peak that no single parameter scale can represent. The same corrections keep most of their effect in the held-out spring–summer evaluation (day bias -8.28 to -3.97 K), so the diagnosed shape is seasonally robust; the profile is interpreted as a diagnosis of the missing process rather than as a correction, and it provides the design basis for the closure of the next
410 subsection.

The optimized profile is a projection of the model–observation residual onto a deliberately chosen, ten-dimensional forcing-side control space, and it is compensatory by construction: it shows how the forcing would have to change for the present model to reproduce the observed surface, not how the forcing actually errs. In particular, the profile is not evidence of a several-kelvin defect in the CMFD air temperature: a literal forcing-error reading would require a station-fused product, whose daily-scale
415 temperatures are well validated (He et al., 2020), to carry a sign-changing sub-daily bias of about 5 K amplitude within the daytime hours, which is implausible. Whether CMFD carries any such sub-daily error is immaterial to the argument and is left to dedicated validation against station observations. Whether the diagnosed deficit originates in the forcing data, in the one-way offline coupling, which permits no two-way adjustment between the land surface and the near-surface atmosphere, or in the land-side exchange formulation cannot be separated within this framework, and none of the conclusions below depends on
420 resolving that question. What the probe does establish is threefold: the channel through which the daytime error is transmitted most strongly (the exchange air temperature; Appendix C4); the functional form the correction must have (a sign change within the daytime hours, which no time-constant parameter in any control space can express); and, through the wind scale settling below unity at 0.72 , internal evidence that the offline configuration couples the surface too strongly to the imposed atmosphere.

Two features of the attribution explain why the closure of the next subsection acts on the exchange air temperature rather
425 than on the other forcings. First, the leverage is not shared evenly: the finite-difference probe identifies the daytime overpass air temperature and the shortwave as the dominant levers, while the wind scale is far weaker. Second, the wind and shortwave corrections are not separable from the air-temperature one: over the optimization the wind scale settles at 0.72 while the shortwave scale stays at its initial value of 1.0 , so the radiative correction is redundant once the air-temperature profile is free to move. The air-temperature profile, by contrast, yields a unique and physically interpretable shape, the missing near-surface
430 diurnal amplitude, and the radiative gain it implicitly requires is energetically equivalent to a shortwave correction. Acting on

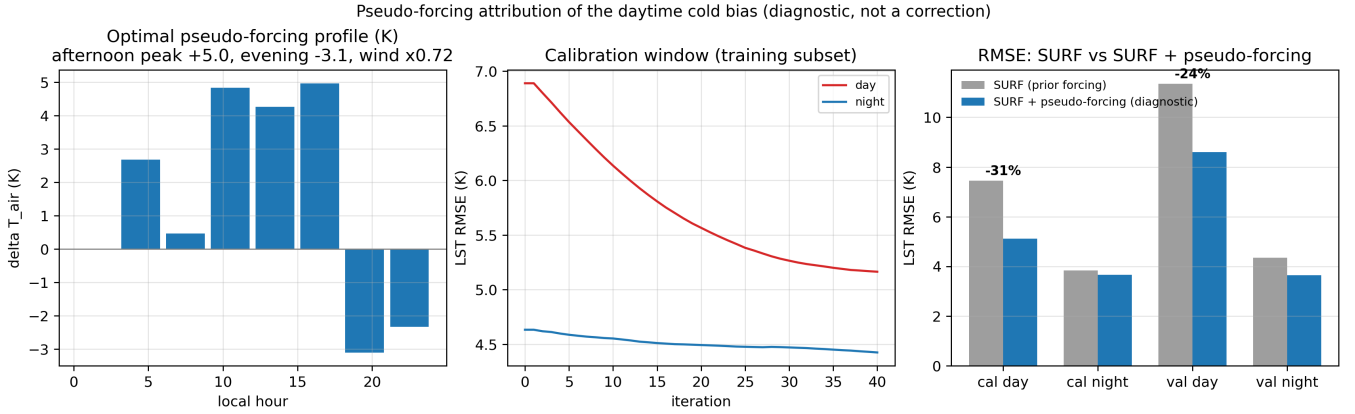


Figure 11. Pseudo-forcing attribution of the daytime cold bias (diagnostic, not a parameterization). Left: the optimized additive air-temperature profile by local 3-hour bin (bars; afternoon peak +5.0 K, evening −3.1 K, wind $\times 0.72$, shortwave unmoved); the profile expresses, in forcing units, the correction the observations require of the near-surface diurnal amplitude of the offline exchange (about 8.1 K peak-to-peak); it is not an estimate of forcing-data error. Centre: day and night LST RMSE over the calibration-window training subset during the ten-control optimization. Right: full-domain LST RMSE with prior forcing versus with the pseudo-forcing applied (SURF vs SURF + pseudo-forcing), in the calibration and held-out windows; the corrections retain most of their effect in the held-out season under the anchored-day protocol (day RMSE −24 %), and the profile is used only as a diagnosis of the missing near-surface diurnal amplitude, not as a model configuration.

the exchange air temperature therefore captures the dominant, unambiguous part of the diagnosed deficit, while the wind part is largely represented by the same amplitude correction. An independent single-channel sensitivity check confirms this ordering: the exchange air temperature drives daytime LST an order of magnitude more strongly than every other forcing channel (five times the next strongest, thirty times the shortwave; Appendix C4), so it is the channel through which the daytime deficit is transmitted most strongly.

5.2 Observation-constrained design of a finite-capacity surface-layer closure

We therefore complement parameter calibration with a process-level representation of the unresolved near-surface coupling. Because the probe cannot separate forcing error from missing coupling physics, the closure is designed as a parameterization of the missing process, not as a correction of the forcing data.

The closure interprets the reanalysis air temperature as a forcing anchored above a finite-capacity surface layer, whose anomaly ΔT tracks the quasi-steady exchange offset $-\lambda_s H$ under the surface sensible heat flux H (downward positive, Fig. 12):

$$\frac{d\Delta T}{dt} = \frac{-\lambda_s H - \Delta T}{\tau(t)}, \quad \tau(t)^{-1} = \frac{w}{\tau_{\text{day}}} + \frac{1-w}{\tau_{\text{night}}}, \quad w = \frac{S}{S + 50 \text{ W m}^{-2}}, \quad (15)$$

where S is the surface downward shortwave of the step, so the day–night transition follows sunrise, sunset, and cloudiness rather than a clock. Discretized explicitly per physics step, $\Delta T_{n+1} = \beta_n \Delta T_n + (1 - \beta_n)(-\lambda_s H_n)$ with $\beta_n = \exp(-\Delta t / \tau_n)$,

Where the surface-layer closure sits: a driver-level correction to the offline atmosphere-surface boundary

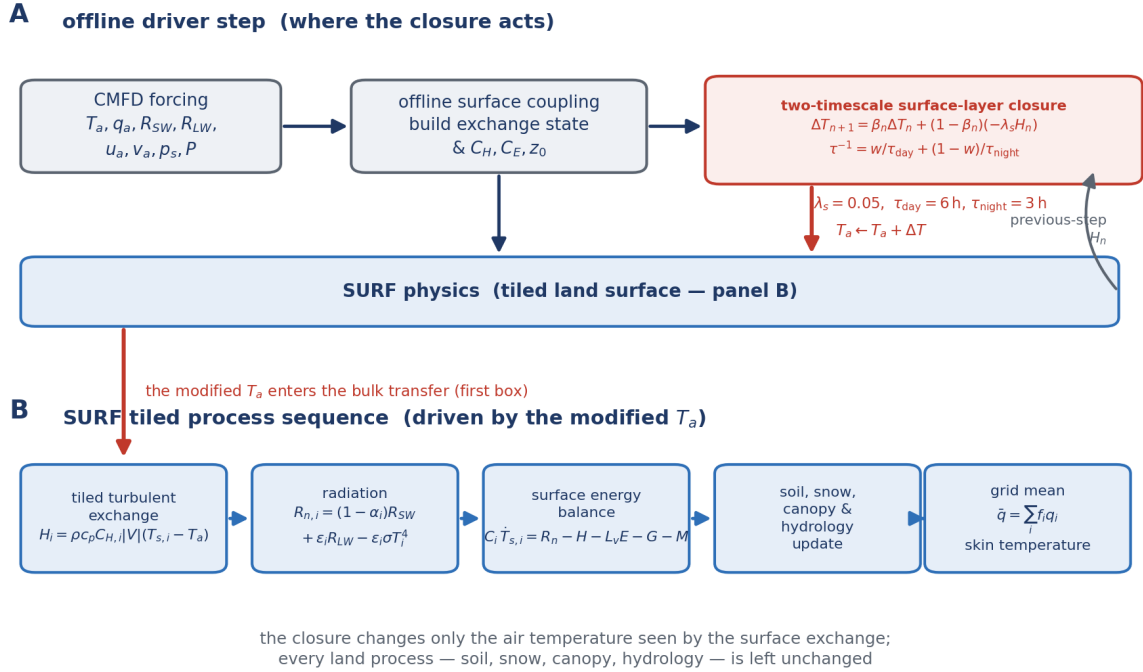


Figure 12. Where the surface-layer closure sits in the offline model. Panel A: the driver step. The closure modifies only the exchange air temperature T_a after the offline surface coupling is built and before the SURF physics is called, a driver-level correction to the atmosphere-surface boundary, not a new land process. Panel B: the tiled SURF process sequence that the modified T_a then drives; every land process (soil, snow, canopy, hydrology) is left unchanged.

and applied to the exchange air temperature only, the closure has three constants: the coupling coefficient λ_s sets the amplitude of the correction, and the two relaxation times set how fast the surface layer tracks it (an effective depth $h = \tau / (\rho c_p \lambda_s)$). Two properties matter. First, the relaxation term is not optional: the undamped quasi-steady form $\Delta T = -\lambda_s H$ is a positive feedback under nocturnal stable stratification, and the first version of this experiment demonstrated the consequence numerically: the integration produces non-finite states at $\lambda_s \approx 0.03 \text{ K (W m}^{-2}\text{)}^{-1}$. Second, $\lambda_s = 0$ reproduces the baseline bitwise, so the closure changes nothing until it is calibrated.

The calibration protocol mirrors Appendix C and is stated here in full. The training windows are ten autumn-winter days of 2018 (days of year 271, 281, ..., 361) and the held-out windows ten spring-summer days (days of year 91, 101, ..., 181); no spring-summer state ever enters a training loss. For every window the land state is restored from the daily archive

455 of the autonomous, unclosed 2018 integration of Sect. 5 at the preceding day boundary; the closure is switched on during a
 one-day (48-step) build-up integrated without gradients, so that its memory state equilibrates before the scored window, and
 the following 48-step forecast day carries gradients. The forcing throughout is the same CMFD 2018 three-hourly field set,
 interpolated to the 1800 s physics step, as the autonomous run (Sect. 3). The loss is the $\cos \phi$ -weighted day-plus-night mean-
 460 squared strict-QC LST residual over every fifth CMFD column (19,542 of 97,709). Within the recurrence, the flux entering the
 quasi-steady target is treated as constant per step, so gradients propagate through the anomaly recursion and its direct effect on
 the exchange air temperature, not through the flux feedback. Stage one trains the coupling coefficient itself (Adam, learning
 rate 0.004 from an initial value of $10^{-3} \text{ K (W m}^{-2})^{-1}$, 40 iterations); stage two optimizes the two timescales jointly by Adam
 on their logarithms (learning rate 0.05, 40 iterations). In both stages an iteration whose gradients are not finite is rejected and
 leaves the parameters at their last stable value. The three constants are calibrated in two stages on the same ten autumn–winter
 465 days. In the first stage the two timescales are tied through a fixed effective depth ($h_s = 600 \text{ m}$, i.e. $\tau_s = \lambda_s \rho c_p h_s$) and the
 single free parameter λ_s settles at its upper bound of $0.05 \text{ K (W m}^{-2})^{-1}$ ($\tau_s \approx 10 \text{ h}$): the data ask for the strongest admissible
 coupling, and the bound-seeking behaviour locates the remaining error in how quickly the layer tracks that coupling. In the
 second stage $\lambda_s = 0.05$ is held fixed and the two relaxation times are trained by the same gradient machinery, the identical
 Adam calibration, loss, and strided column subset, with the times parameterized in log space and initially bounded to $[2, 12] \text{ h}$.
 470 Started from the tied value of 10.1 h for both, the optimizer separates the timescales within about 30 accepted steps and becomes
 stationary at $\tau_{\text{day}} = 2.47 \text{ h}$ and $\tau_{\text{night}} = 2.36 \text{ h}$ (training loss 481 to 401 on the calibration subset); the trailing iterations are
 then rejected by the non-finite-gradient guard, because the optimizer has walked to the edge of its own integrable range. That
 endpoint is unusable. Verified step by step in state space rather than at observation times, the held-out day-131 forecast diverges
 in two clusters of Tibetan-Plateau columns ($30\text{--}31^\circ\text{N}$, $84\text{--}86^\circ\text{E}$ and $37\text{--}38^\circ\text{N}$, $87\text{--}88^\circ\text{E}$): there the grid-mean sensible heat flux
 475 sustains an intrinsic two-step ($\sim 1 \text{ h}$) limit cycle even at vanishing anomaly, and near the morning transition its sensitivity to the
 exchange-air anomaly is large enough that the closure’s quasi-steady target $-\lambda_s H$ reinforces the oscillation instead of damping
 it; the anomaly passes 10^3 K within a dozen steps of the dawn transition, and the first non-finite state appears at forecast step 56
 (Appendix C). Observation-space verification cannot see this: the strict MODIS quality control removes the divergent columns
 from the sampled statistics, whose window aggregates stay clean. The acceptance test for a learned closure must therefore be
 480 state-space verification, applied here to the full sweep grid over all 20 anchored evaluation days. The verified set is a resonance
 structure rather than a rectangle: $\tau_{\text{night}} = 2 \text{ h}$ exceeds 10^3 K on at least one anchored day at $\tau_{\text{day}} \in \{5, 8, 10\} \text{ h}$ and stays finite
 but elevated at $\tau_{\text{day}} = 6 \text{ h}$ (worst transient 150 K), $\tau_{\text{day}} = 5 \text{ h}$ diverges on day 121 at every τ_{night} except 3 h, and ($\tau_{\text{day}} = 6 \text{ h}$,
 $\tau_{\text{night}} = 5 \text{ h}$) diverges on day 121, whereas ($\tau_{\text{day}} = 6 \text{ h}$, $\tau_{\text{night}} \in \{3, 4\} \text{ h}$) and $\tau_{\text{day}} \in \{8, 10\} \text{ h}$ at $\tau_{\text{night}} \in \{3, 4, 5\} \text{ h}$ pass all
 20 days with no column beyond 141 K (worst single-column transient 109 K for the adopted combination, decaying, against
 485 49 K for the tied closure). Resetting the training bounds in light of this test and repeating the gradient calibration from the same
 tied start, the optimizer again presses toward the fastest admissible tracking and settles at the verified corner, $\tau_{\text{day}} = 6 \text{ h}$ and
 $\tau_{\text{night}} = 3 \text{ h}$ (effective depths ≈ 360 and 180 m ; training loss 433, an 8 % premium over the divergent optimum that is the price
 of integrability). The calibrated configuration is hereafter referred to as SURF-Opt; the uncalibrated model remains SURF.

Table 6 gives the full-domain verification of SURF-Opt against the uncalibrated SURF. The two-timescale closure removes 67 % of the daytime bias in the calibration window (-5.48 to -1.83 K; RMSE 7.45 to 4.82 K) and 84 % in the held-out spring–summer window (-8.28 to -1.35 K; RMSE 11.34 to 7.17 K), seven to thirteen times the bias removal of the eight table scales of Appendix C. Unlike the pseudo-forcing profile, which is an additive adjustment fitted to one year’s forcing, the closure is a process formulation: it is defined by three physical constants, carries no per-year information, and its improvement is as large in the window it never saw. The night passes quantify the trade-off. The calibration-window night pass slightly improves (-0.43 to -0.31 K; RMSE 3.83 to 3.66 K); the held-out night bias warms from $+1.07$ to $+2.08$ K at essentially unchanged RMSE (4.35 to 4.48 K): the daytime heat no longer lost to the atmosphere is stored in the soil and returned after sunset. What remains of the daytime bias is a modest cold offset (-1.4 K) in the held-out window; the warm nights and the shortwave factor of the attribution experiment point at the same next structural target, a radiative or albedo deficit that a surface-layer closure cannot supply. Two caveats accompany these gains. First, like the probe that motivated it, the closure is compensatory in character: it represents, in three constants, whatever unresolved physics projects onto the exchange-temperature channel, most plausibly the missing two-way coupling between the surface and the near-surface atmosphere, an interpretation consistent with the implied effective depths (360 and 180 m), which are of the order of the diurnal heat capacity of the atmospheric surface layer. Its credibility therefore rests on the held-out transfer reported here rather than on a first-principles derivation. Second, the constants are calibrated against MODIS LST under a single forcing (CMFD) and are tied to this model–forcing combination; repeating the calibration under an alternative forcing and validating the sub-daily structure of the exchange level against station observations are the decisive next tests. Broader parameter calibration, including heterogeneous static fields and slow soil-state memory, remains complementary to, not superseded by, this closure.

Figures 14 and 15 resolve the improvement spatially and day by day, and the monthly RMSE curves of Fig. 16 extend it to the full seasonal cycle. The SURF daytime bias is strongest over the arid northwest and the Tibetan Plateau and weakest over southern and northeastern China, and the closure acts exactly where the bias lives: the box-mean day bias over the northwestern deserts ($75\text{--}105^\circ\text{E}$, $35\text{--}48^\circ\text{N}$) improves from -10.8 to -2.9 K in the held-out window and over the Tibetan Plateau from -9.4 to -2.9 K, while northeastern China shifts from -3.1 to $+2.1$ K. Across all evaluated columns the absolute daytime bias decreases at 70–76 % of columns, and the fraction of columns biased by more than 10 K drops from 16 % to 2 % (calibration) and from 37 % to 10 % (held-out). The one systematic cost is visible in the same maps: southern China, nearly unbiased in SURF, warms by about $+3$ K, the regional signature of the night-pass warming already quantified in Table 6. The per-day panels show the reduction is not an average artefact: on every one of the 20 evaluation days, in both windows, the SURF-Opt day bias is warmer and the day RMSE lower than SURF. The daily curves of Fig. 17 show the outcome at daily resolution: in the day pass, SURF-Opt tracks not only the observed seasonal cycle but the day-to-day variability through the whole year, whereas SURF sits several kelvin low in the warm season; the night-pass panel exposes the warm-season night warming directly (about $+2$ K in July). The monthly RMSE curves of Fig. 16 extend the check to the full seasonal cycle, from continuous year-long integrations: the day-pass RMSE decreases in every month of the year (by 0.9–4.4 K, largest in May), while the night-pass cost is confined to May–August (at most 0.6 K, in 4 of 12 months); the cold-season night pass is unchanged to slightly improved.

Table 6. Full-domain MODIS LST verification of SURF-Opt against the uncalibrated spun-up SURF. “SURF-Opt” is the optimized configuration of Eq. (15) with $\lambda_s = 0.05 \text{ K (W m}^{-2}\text{)}^{-1}$ and gradient-calibrated relaxation times ($\tau_{\text{day}} = 6 \text{ h}$, $\tau_{\text{night}} = 3 \text{ h}$, on the state-space-verified stability boundary) obtained by the two-stage calibration on the ten autumn–winter days. The diagnostic experiments of Appendix C are not model configurations and are therefore not tabulated: the eight-control table-scale calibration removes at most 0.5 K of the daytime bias within that tested control space, and the ten-control pseudo-forcing attribution removes most of the calibration-window day bias (-5.48 to -1.44 K , RMSE 7.45 to 5.11 K) and retains most of that reduction in the held-out window (day bias -8.28 to -3.97 K).

Window	Pass	SURF		SURF-Opt	
		bias (K)	RMSE (K)	bias (K)	RMSE (K)
Calibration (autumn–winter)	day	−5.48	7.45	−1.83	4.82
	night	−0.43	3.83	−0.31	3.66
Held-out (spring–summer)	day	−8.28	11.34	−1.35	7.17
	night	+1.07	4.35	+2.08	4.48

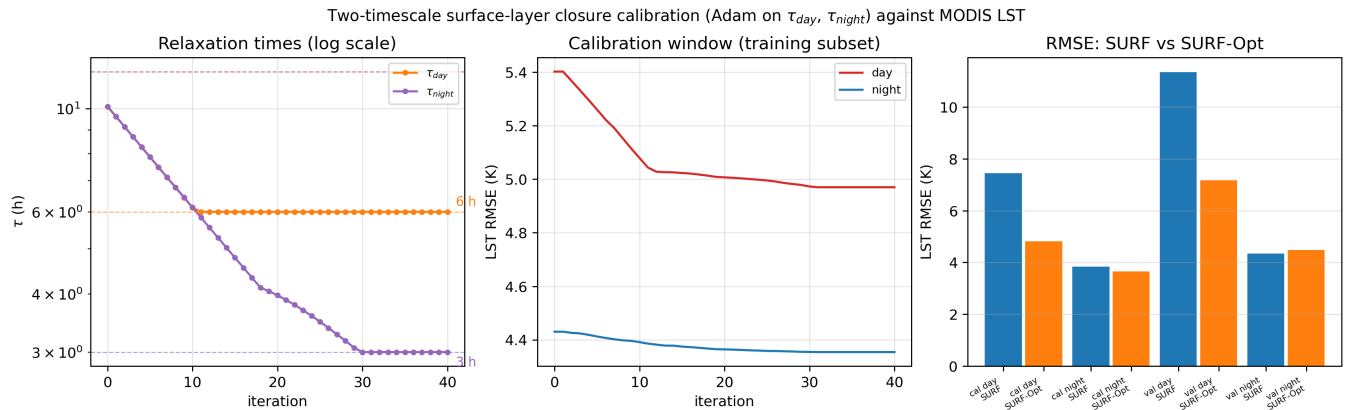


Figure 13. Two-timescale surface-layer closure calibration (Eq. (15)). Left: trajectories of the two relaxation times under Adam on $(\log \tau_{\text{day}}, \log \tau_{\text{night}})$, started from the tied value of 10.1 h and settling at the corner of the state-space-verified set (dashed: the training bounds $6 \leq \tau_{\text{day}} \leq 12 \text{ h}$ and $3 \leq \tau_{\text{night}} \leq 12 \text{ h}$; the unconstrained endpoint at (2.47, 2.36) h lies outside it and diverges). Centre: calibration-window day and night LST RMSE over the training subset. Right: SURF-versus-SURF-Opt full-domain RMSE in the calibration and held-out windows: the day-pass improvement transfers to the window never used in training.

Because the closure acts only on the exchange-level air temperature and the grid-mean sensible heat flux, physical quantities computed identically in either implementation, it requires no differentiable infrastructure to apply: the identical recurrence
525 has been implemented at the driver level of the Fortran offline model, where the compiled-in default $\lambda_s = 0$ reproduces the unpatched driver bitwise (Appendix D).

Daytime LST bias vs MODIS: SURF vs SURF-Opt (two-timescale surface-layer closure, $\lambda_s=0.05 \text{ K/(W m}^{-2}\text{)}$, $\tau_{\text{day}}=6 \text{ h}$, $\tau_{\text{night}}=3 \text{ h}$)

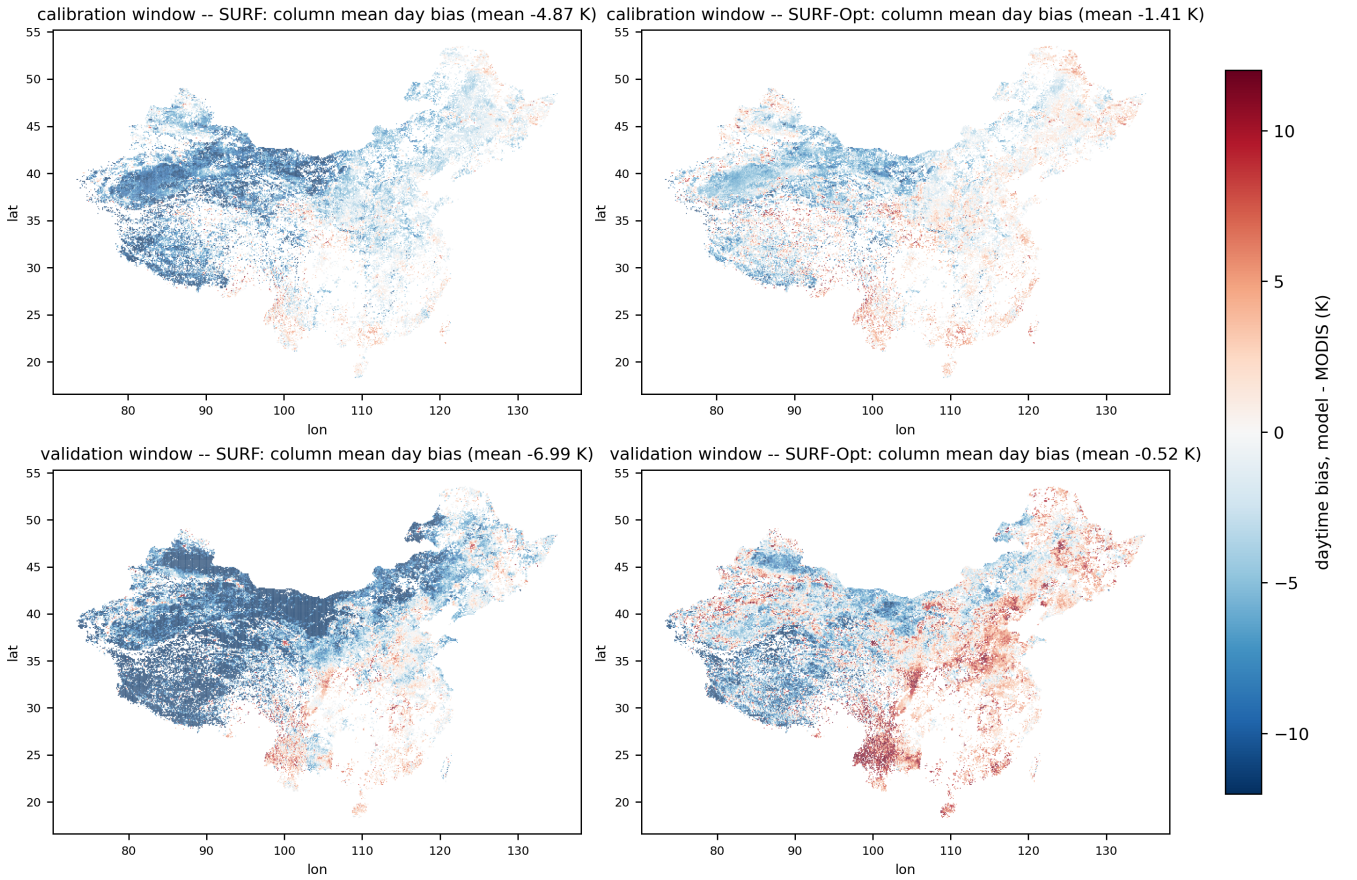


Figure 14. Column-mean daytime LST bias against MODIS for SURF (left) and SURF-Opt (right), in the calibration window (top row) and the held-out spring–summer window (bottom row). Panel titles give the area-weighted window mean; all panels share a divergent colour scale ($\pm 12 \text{ K}$). The closure acts where the bias lives: the strong cold biases over the arid northwest and the Tibetan Plateau are largely removed, while nearly unbiased southern China warms by about $+3 \text{ K}$, the regional signature of the night-pass warming quantified in Table 6.

6 Computational performance

The tensor formulation exposes the independent SURF columns to batched accelerator execution without changing the prognostic calculation. We measured the autonomous offline implementation against the Fortran reference over progressively larger subsets of the effective 2018 CMFD domain. Each run used the same restart, forcing representation, double-precision state and arithmetic, and 1800 s physics step as Sect. 3.2. A timed segment contains 240 continuous steps (5 simulated days). The Fortran reference was timed with `osmMASTER` under 32 OpenMP threads on an Intel Xeon Gold 6430 host; the accelerator experiments use one NVIDIA A100 80 GB GPU. Each configuration was run independently three times and the median is reported. All timing runs use the pinned software environment of Sect. 3.2.

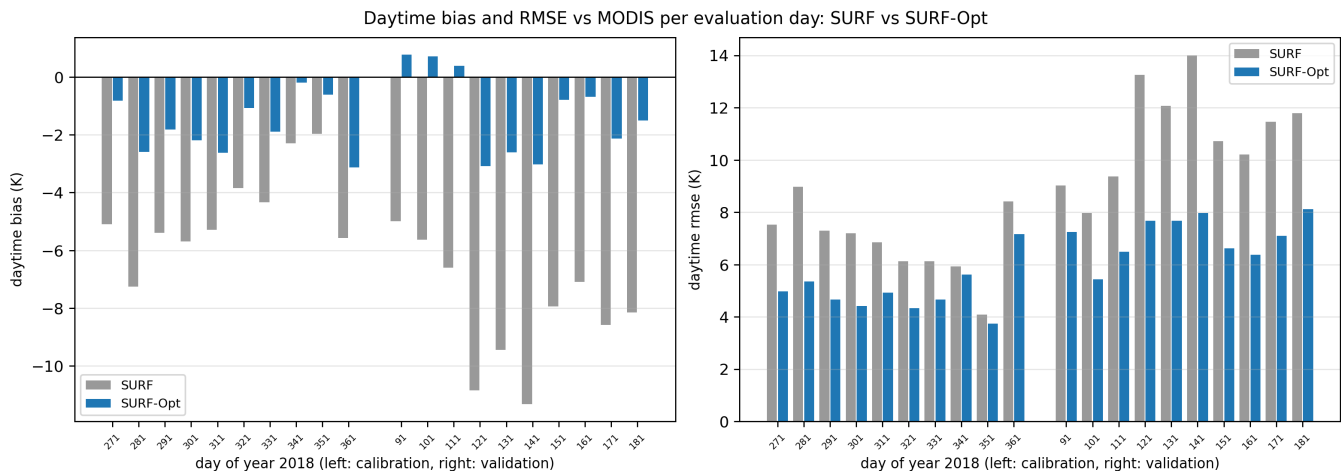


Figure 15. Per-evaluation-day, area-weighted daytime bias (left) and RMSE (right) against MODIS for SURF (grey) and SURF-Opt (blue), over the ten calibration days (days of year 271–361) and the ten held-out spring–summer days (days of year 91–181).

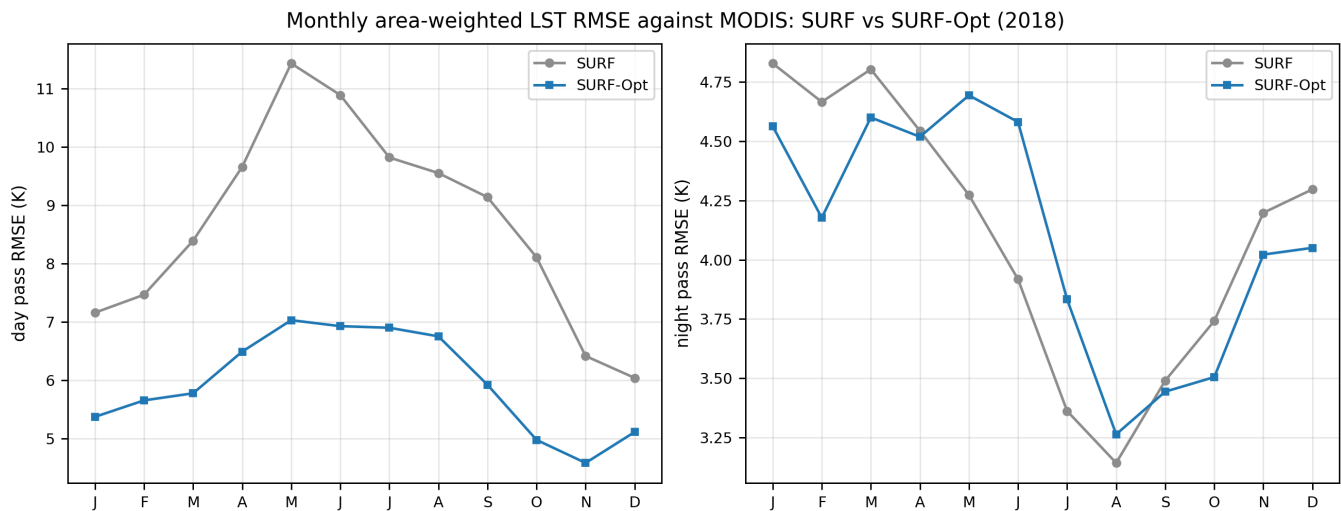


Figure 16. Monthly, area-weighted LST RMSE against MODIS (strict QC) for the SURF (grey) and SURF-Opt (blue), day pass (left) and night pass (right), from continuous full-year 2018 integrations of the torch implementation under the observation operator of Sect. 5. The day-pass RMSE is lower in every month; the night-pass degradation is confined to March–September.

535 The timing boundary is deliberately narrow. State output is disabled, the optional per-step host-side non-finite check is disabled, and the GPU stream is synchronized immediately after the final model step. The final state is then checked for finite values outside the timed interval. Consequently, integration time measures the continuous physics calculation rather than initial-state loading, diagnostic transfer, file serialization, or the one-time forcing load; the Fortran timing is the internal time-step-loop section, excluding initialization, forcing reads, and restart I/O.

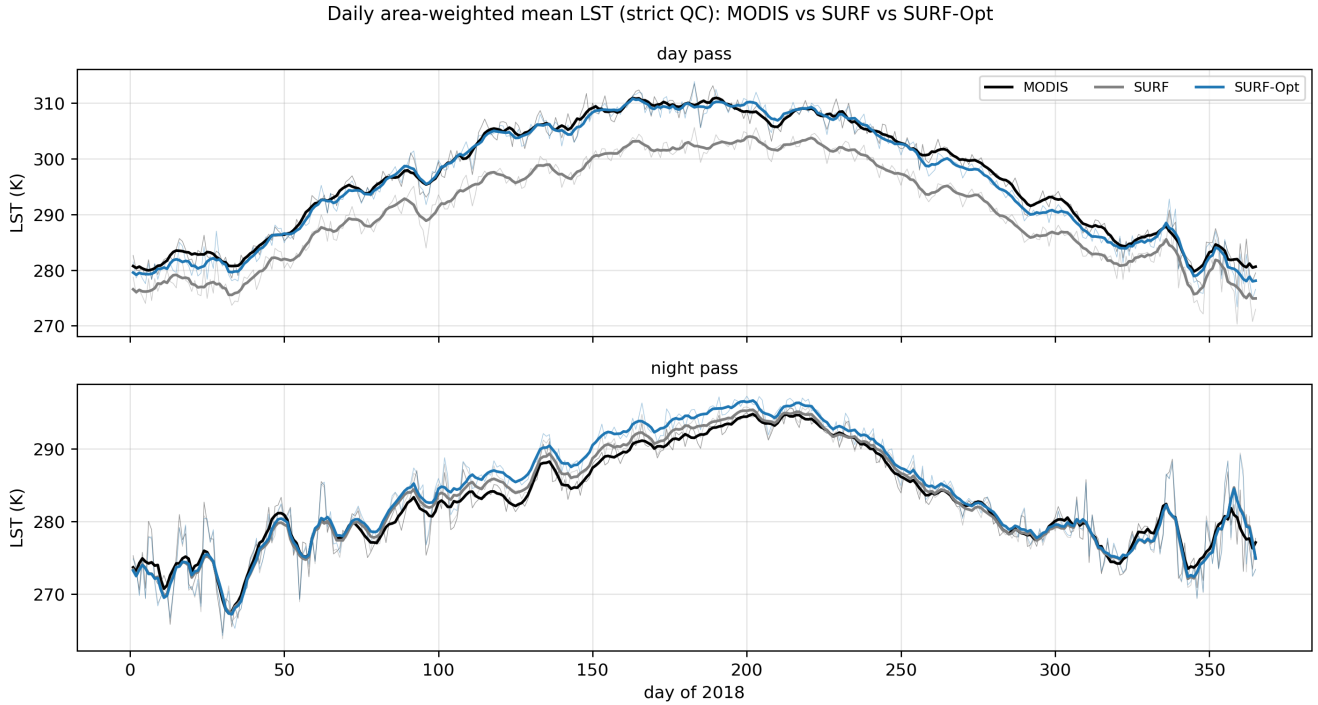


Figure 17. Daily, area-weighted mean LST from the same continuous year-long integrations (strict QC): MODIS (black), SURF (grey), and SURF-Opt (blue), for the day pass (top) and the night pass (bottom). Thin lines are the daily values, thick lines a seven-day running mean. In the day pass, SURF-Opt tracks the observed seasonal cycle and its day-to-day variability through the whole year, whereas SURF sits several kelvin low in the warm season; the night-pass panel exposes the warm-season night warming directly (about +2 K in July). The monthly-mean view of the SURF forecast against the same observations is given in Fig. 10.

Two accelerator configurations are reported. The eager configuration launches each tensor operation individually; the CUDA-graph configuration records one complete physics step (forcing construction plus the full model update) into a single replayable graph, so subsequent steps launch once. Making the step capturable required eliminating every host synchronization in the step: scalar reads of device tensors were replaced by broadcast views, per-step host-side constant materializations were cached on the device, and the one state-dependent host branch was replaced by an idempotent masked update. Each change is bit-identical to the eager path, and the captured integration reproduces the eager trajectory bit for bit at 1,000 and at 97,709 columns.

Figure 18 shows the three configurations. The Fortran reference is the most efficient configuration below about ten thousand columns, where per-step kernel-launch overhead dominates the Torch integrations: the step issues thousands of small CUDA kernels, and at small batches their launch latency exceeds the computation they carry. Recording the step once and replaying it removes that floor: the graph configuration outperforms the Fortran reference from about eight thousand columns upward, and reaches $3.84\times$ over the complete effective domain. For the complete effective CMFD domain, the graph configuration reaches 1.69×10^6 column-steps s^{-1} , compared with 4.39×10^5 column-steps s^{-1} for the Fortran reference. The gain comes

Table 7. Batch-size performance of the double-precision autonomous offline SURF integration against the Fortran reference on the same columns. Values are median wall-clock integration times in seconds (parentheses give the minimum–maximum range across three repeats). Each integration contains 240 continuous 1800 s steps. The rightmost columns give each accelerator configuration’s speed-up over the 32-thread Fortran reference.

Active columns	Fortran 32 threads (s)	Torch A100 eager (s)	Torch A100 graph (s)	speed-up over Fortran eager	speed-up over Fortran graph
1,000	0.55 (0.53–0.61)	16.98 (16.32–17.71)	4.60 (4.59–4.63)	0.03×	0.12×
10,000	8.29 (8.27–8.66)	15.81 (15.39–16.11)	5.09 (5.09–5.13)	0.52×	1.63×
50,000	28.26 (28.14–28.68)	14.87 (14.62–14.98)	8.37 (8.35–8.43)	1.90×	3.38×
97,709	53.38 (49.60–53.57)	18.13 (18.12–18.67)	13.90 (13.81–13.93)	2.94×	3.84×

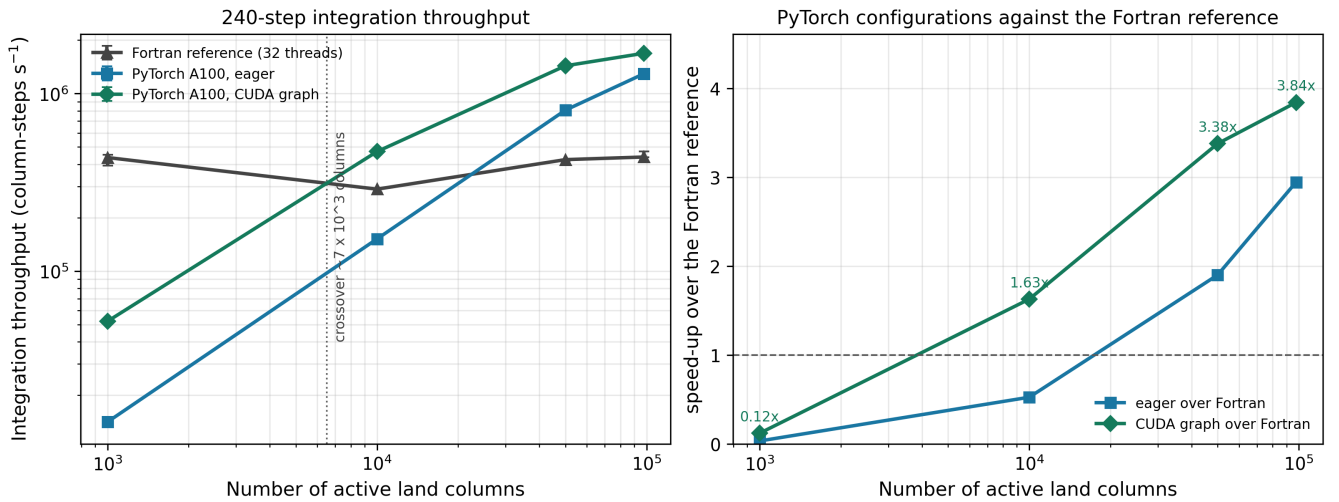


Figure 18. Integration throughput and accelerator speed-up for the autonomous offline SURF implementation over 240 continuous steps, against the Fortran reference on the same columns. Left: throughput for the Fortran reference and the A100 eager and CUDA-graph configurations; points are repeat medians, error bars span the observed minimum–maximum integration times. Right: each accelerator configuration’s speed-up over the Fortran reference.

from batched evaluation of the same 1800 s double-precision process sequence, rather than from an altered time step or forcing representation.

These gains are in line with published ports of comparable double-precision physics. The GT4Py rewrite of the FV3 dynamical core reports 3.3× over its optimized Fortran reference on float64 (Dahm et al., 2023); a JAX port of the multilayer-canopy CLM-ml reports 2.2×–4.7× over sequential Fortran, with no benefit from single precision on this bandwidth-bound class of code (Lahlou et al., 2026); and the full OpenACC port of ICON reports 5.5× socket-to-socket for the complete model (Lapillonne et al., 2026). CUDA-graph recording of small soil and turbulence kernels in the ICON port translated to about 5 %

end-to-end; the larger gain here follows from the unusually high kernel count per step of the tiled column physics. Double-digit GPU-versus-CPU claims in this literature trace to unoptimized CPU baselines or many-GPU against single-CPU-node comparisons, and are not the like-for-like regime measured here.

7 Discussion

The controlled boundary-driven integration isolates the numerical translation of the SURF process sequence from the construction of atmospheric lower-boundary quantities. The annual autonomous experiment complements that test by exercising the standalone roughness and surface-layer exchange closure under a realistic, continuously varying forcing record. These experiments are complementary: the 30-day case tests the translated SURF process sequence under fixed boundaries, whereas the annual case tests the complete offline pathway. Over a long nonlinear integration, floating-point round-off can be amplified when an evolving column approaches a thresholded snow, wetness, or surface-layer regime. The small annual domain-wide errors nevertheless show that such perturbations remain limited for the evaluated CMFD trajectory.

The offline closure is required only when SURF is driven outside an atmospheric model. In a coupled OpenIFS application, the roughness and surface-layer exchange quantities are supplied through the atmospheric-physics interface, so the translated SURF process sequence uses the corresponding external-boundary path. This separation makes the reported verification relevant both to a standalone land-surface workflow and to the coupled application for which SURF was originally designed.

Several qualifications define the scope of these results. The Fortran–Torch CMFD experiment tests numerical consistency, whereas the MODIS comparison provides an independent but configuration-dependent evaluation of surface temperature and snow cover. The initial state, static fields, forcing interpolation, and snow configuration are shared by the two implementations. Agreement therefore demonstrates consistency of the numerical pathway under those conditions, not uniqueness or physical optimality of the state. The annual autonomous comparison also contains the independently reconstructed offline surface-layer closure, so its small residuals combine round-off effects, threshold activation, and any closure-level differences. For this reason, the controlled boundary-driven integration is the primary evidence for the SURF translation, while the autonomous integration is evidence for stability of the complete offline workflow. The annual water budget closes at numerical precision. The reported surface-energy residual is instead a diagnostic of the staged discrete solve and is not interpreted as a prognostic conservation residual. The MODIS daytime contrast and snow-cover underestimation show that numerical agreement does not remove forcing, initialization, or parameterization error. A longer multi-year protocol should use seasonally matched land analyses and additional in situ water and energy flux observations.

The differentiable translation and a learned emulator address the same interface from opposite directions. The aiLand v1 emulator of the companion ecLand scheme (Raoult et al., 2026) approximates the IFS land-surface lineage with a multi-layer perceptron: pre-trained on ecLand offline integrations, it reproduces the prognostic soil state with a 90-day RMSE of 1.19 K, remains stable over continuous four-year autoregressive integrations, and is then fine-tuned against FLUXNET flux observations, reducing latent and sensible heat-flux RMSE by 30 and 20 % at independent towers. The derivative of such a surrogate, however, is the derivative of a one-kelvin approximation of the physics, not of the physics itself, and its observational

gains are absorbed into network weights that map back to neither physical parameters nor the process formulation. The path taken here is complementary: derivatives point at the operational physics, so observation constraints land on physical parameter tables and on the design of the surface-layer closure, and the resulting correction ports back to the Fortran reference without
595 re-tuning. Emulators and differentiable physics translations are natural partners: the former for inexpensive ensembles and assimilation acceleration, the latter for mechanism diagnosis, parameter estimation, and structural correction.

The observation-space workflow of Sect. 5 closes the loop between diagnosis and process change. The gradient-based experiments of Appendix C show that automatic differentiation supplies efficient derivatives, but experiment design and seasonal state diversity determine whether those derivatives carry independent parameter information; the calibration then quantifies the
600 parameter-explained fraction of the misfit (about 0.5 K of the 5.5–8.3 K daytime cold offset) and attributes the remainder to the missing near-surface diurnal amplitude of the offline exchange, which the two-timescale surface-layer closure largely supplies. The bound holds for a specific control space: it covers neither heterogeneous static fields such as soil-texture, hydraulic, and albedo maps nor the slow soil-water memory that the anchored ten-day windows suppress, so parameter calibration over a richer control space remains complementary to, not superseded by, the closure. The pseudo-forcing profile behind the closure
605 design is likewise a compensatory projection of the residual onto a forcing-side control space, not evidence of a several-kelvin error in the station-fused forcing product; what it establishes is the functional form of the deficit, an amplitude whose sign changes within the day and which no time-constant parameter can express, together with the channel through which it acts. The closure constants are calibrated against one forcing product and one observation stream, and repeating the diagnosis and calibration under an independent forcing is the natural next step. The differentiable model is therefore most valuable for what
610 it can separate: the parameter-explained fraction of an observation misfit from its structural remainder, and the physical shape of that remainder.

The Fortran-guided retrieval (Appendix B) is the state-estimation counterpart of the parameter workflow: an inverse calculation whose target is produced by a separately executed reference model and whose later state is withheld from the update.

8 Conclusions

615 The PyTorch implementation reproduces the translated OpenIFS SURF process sequence in a controlled full-domain boundary-driven integration and is closely aligned with the autonomous Fortran offline solution through a one-year, 97,709-column CMFD integration. The equivalence persists over a ten-year successively restarted integration (2009–2018) without error growth. Within a pinned software environment, repeated PyTorch integrations are bitwise reproducible; cross-build round-off differences, amplified by a decadal offline integration, motivate recording the interpreter, library builds, and source hashes
620 in every experiment manifest. CUDA-graph execution of the batched integration reaches 3.8-fold the 32-thread Fortran reference integration time for the complete effective CMFD domain, a range consistent with published double-precision ports of comparable physics (Dahm et al., 2023; Lahlou et al., 2026; Lapillonne et al., 2026). The annual water budget closes with an RMS residual of $5.90 \times 10^{-12} \text{ kg m}^{-2}$ across the evaluated columns. Automatic derivatives through a 24 h trajectory agree with centred finite differences to within 4×10^{-4} relative error for representative forcing, state, and vegetation controls, with every

625 conditional regime switch retained unsmoothed: the differentiated operator is the operator whose equivalence is verified, and the threshold-sensitive columns where a derivative becomes a local linearization are mapped and reported rather than hidden. The resulting model provides a numerically verified and accelerator-ready SURF implementation for offline process studies; in coupled applications, atmospheric lower-boundary states are supplied through the established OpenIFS interface.

Independent MODIS evaluation separates this numerical verification from physical performance. The evaluated offline con-
630 figuration captures nighttime LST variability but under-amplifies daytime surface warming and underestimates snow-cover extent. These diagnosed, spatially structured errors provide observation-based targets for the differentiable sensitivity and calibration workflow rather than being attributed to the PyTorch translation. Because the implementation is differentiable end to end, the daytime cold bias can be carried past diagnosis to correction: it is decomposed into a parameter-explained part bounded within the tested control space and a structural remainder, the functional form of the missing near-surface process is
635 identified by reverse-mode attribution, and a new surface-layer closure is calibrated and verified against the same observations. This closing of the loop between satellite observation and process representation is the central capability the differentiable formulation adds to the verified translation.

The gradient-based experiments against the same observations (Appendix C) demonstrate end-to-end parameter training against real satellite data with exact reverse-mode derivatives: an observation-space information analysis gives condition num-
640 bers of 2.96×10^5 in spring–summer and 67 in autumn–winter, an observing-system simulation with the real MODIS operator recovers imposed table scales to within 0.3–1.1 % under 1 K observation noise, and the real-observation calibration reduces day-pass LST RMSE by 5.7 % in the calibration window and 3.8 % in the seasonally held-out evaluation. Their equally important output is the bound itself: within the tested space of eight global table scales, at most about 0.5 K of the 5.5–8.3 K daytime cold offset is parameter-explained: a bound on this control space, not a general verdict on parameter optimization, which over
645 richer or spatially heterogeneous parameter fields remains complementary to the closure below. A ten-control pseudo-forcing attribution shows the remainder corresponds to roughly 8.1 K peak-to-peak of missing near-surface diurnal amplitude, a compensatory profile that states the functional form of the deficit without locating its origin uniquely in the forcing data.

The residual daytime cold offset beyond the calibrated parameter scales is corrected, in large part, by a process change that complements rather than replaces a parameter change. The two-timescale finite-capacity surface-layer closure of Sect. 5.2,
650 yielding the optimized configuration SURF-Opt, is derived from the pseudo-forcing diagnosis, which on its own is a seasonally robust diagnostic profile rather than a parameterization or a correction of the forcing data, and is calibrated on ten days in two gradient stages: the coupling coefficient settles at its admissible bound ($\lambda_s = 0.05 \text{ K (W m}^{-2})^{-1}$), and the day and night relaxation times are trained with the same machinery. The freely trained timescales converge outside the integrable domain; a step-level state-space check shows they resonate with an intrinsic dawn-time sensible-heat limit cycle in a handful of Tibetan-
655 Plateau columns, an instability that the quality-controlled observation scores cannot see, so the training bounds are reset to the state-space-verified domain and the optimizer settles on its boundary at $\tau_{\text{day}} = 6 \text{ h}$ and $\tau_{\text{night}} = 3 \text{ h}$. The calibrated closure removes 67–84 % of the day bias (calibration to held-out; day RMSE 11.34 to 7.17 K), on every one of the 20 evaluation days and in every month of the year, at the cost of a warm-season night warming of about +1 K that leaves the night RMSE essentially unchanged. The closure recurrence is implemented at the driver level of the Fortran reference and verified to be

660 bitwise invisible when inactive (Appendix D); being expressed entirely in the grid sensible heat flux and the exchange air temperature, it applies there without re-tuning, so closures designed in the differentiable implementation are not tied to it.

Using an independent Fortran trajectory, a compact initial-state retrieval also demonstrates that the verified gradients can reduce a deliberately introduced land-state error and improve a later held-out state (Appendix B). This provides a controlled foundation for subsequent observation-constrained land-state and parameter-estimation studies.

665 **Appendix A: Decadal integration: detailed verification**

This appendix collects the detailed verification of the ten-year autonomous CMFD integration (2009–2018) summarized in Sect. 3.4. Both implementations use the same shared initial state; each year is restarted from the terminal state of the previous year, and the final forcing records of each annual file are linked to the first records of the following year, so that the interpolated forcing is continuous at each year boundary. The Fortran reference integrates each year in the offline configuration,
670 restarted from the previous year’s terminal state; the PyTorch implementation carries the state in memory across the same year boundaries. States are archived every ten days and at each annual terminal step.

The offline forcing alignment must be replicated exactly rather than approximately. When a variant of the PyTorch implementation samples the state meteorology at the raw three-hourly records instead of at the linearly interpolated physics-step boundaries (radiation and precipitation held at the next record, as in the reference configuration), the domain-RMSE soil-
675 temperature difference reaches 2.1×10^{-1} K within ten days and isolated snow columns accumulate over 10^2 kg m⁻² of spurious snow water equivalent within the first year, even though the underlying forcing data are identical. All successively restarted results use the exact reference alignment.

Table A1. Terminal day-3652 (31 December 2018) Fortran–PyTorch comparison for the ten-year autonomous CMFD integration. Relative RMSE is normalized by the root-mean-square Fortran field over the stated mask. “All” contains 97,709 columns; the resolved-snow mask (snow-cover fraction above 10^{-3}) contains 82,132 columns at day 3652.

Quantity (unit)	mask	RMSE	relative RMSE (%)
Soil temperature, layer 1 (K)	all	8.05×10^{-4}	3.06×10^{-4}
Soil temperature, layer 2 (K)	all	7.09×10^{-4}	2.67×10^{-4}
Soil temperature, layer 3 (K)	all	6.12×10^{-4}	2.25×10^{-4}
Soil temperature, layer 4 (K)	all	6.81×10^{-4}	2.44×10^{-4}
Soil water, layer 1 (kg m^{-2})	all	4.24×10^{-3}	2.46×10^{-2}
Soil water, layer 2 (kg m^{-2})	all	8.86×10^{-3}	1.66×10^{-2}
Soil water, layer 3 (kg m^{-2})	all	2.45×10^{-2}	1.28×10^{-2}
Soil water, layer 4 (kg m^{-2})	all	6.36×10^{-2}	1.19×10^{-2}
Snow water equivalent (kg m^{-2})	resolved snow	8.39×10^{-3}	2.38×10^{-3}
Snow temperature (K)	resolved snow	1.06×10^{-3}	4.21×10^{-4}
Snow density (kg m^{-3})	resolved snow	1.59×10^{-2}	1.11×10^{-2}
Snow albedo (–)	resolved snow	1.26×10^{-5}	2.18×10^{-3}

The error level of an individual annual terminal is set by the forcing realization rather than by the experiment design. The successively restarted integration starts from the same common spun-up initial state as the one-year experiment of Sect. 3.2, under the 2009 forcing instead of the 2018 forcing: at its first-year terminal (day 365 of 2009) the layer-1 soil-temperature RMSE is 6.1×10^{-4} K, 2.2 times *larger* than the one-year value of Table 2, while the layer-4 soil-water RMSE is 3.3×10^{-2} kg m^{-2} (6.3×10^{-3} % relative), six times *larger* than the one-year value (5.6×10^{-3} kg m^{-2} , 1.0×10^{-3} %). With identical code and initial state, this spread measures the realization dependence of the round-off-seeded branch selection, which phase, density, or stability thresholds a given year’s weather crosses, and it is of the same size as the year-to-year oscillation of Fig. A1 itself. No single annual terminal is therefore “the” translation error of the autonomous configuration; the bounded envelope across the decade is.

Three observations support the branch-event reading summarized in Sect. 3.4. First, each field’s error peaks in its own threshold-active season: layer-1 soil-temperature RMSE peaks in July at 1.5×10^{-2} K, eight times its January minimum; snow-water RMSE peaks in the March melt season nearly two orders of magnitude above its summer minimum; the layer-4 soil water and temperature errors peak in October–November, the lagged integral of the summer evaporation season. Second, the annual-terminal errors rise until the 2012–2014 terminals and then decline (a $\log_{10}$ slope of -0.2 to -0.5 per decade), and each year’s terminal error correlates with that year’s persistent snow-cover extent ($r = +0.82$ for layer-4 soil temperature and $r = +0.75$ for layer-4 soil water against the annual snow-covered column–days), so the peak years are those with the most persistent snow cover. Third, the columns carrying the largest layer-1 soil-temperature error are mostly snow-free (median snow water 0.0 kg m^{-2} ; snow involved in 29 % of records), which locates the summer maxima in soil-moisture branch activity rather than in snow.

Decadal Fortran-PyTorch terminal errors, 2009–2018

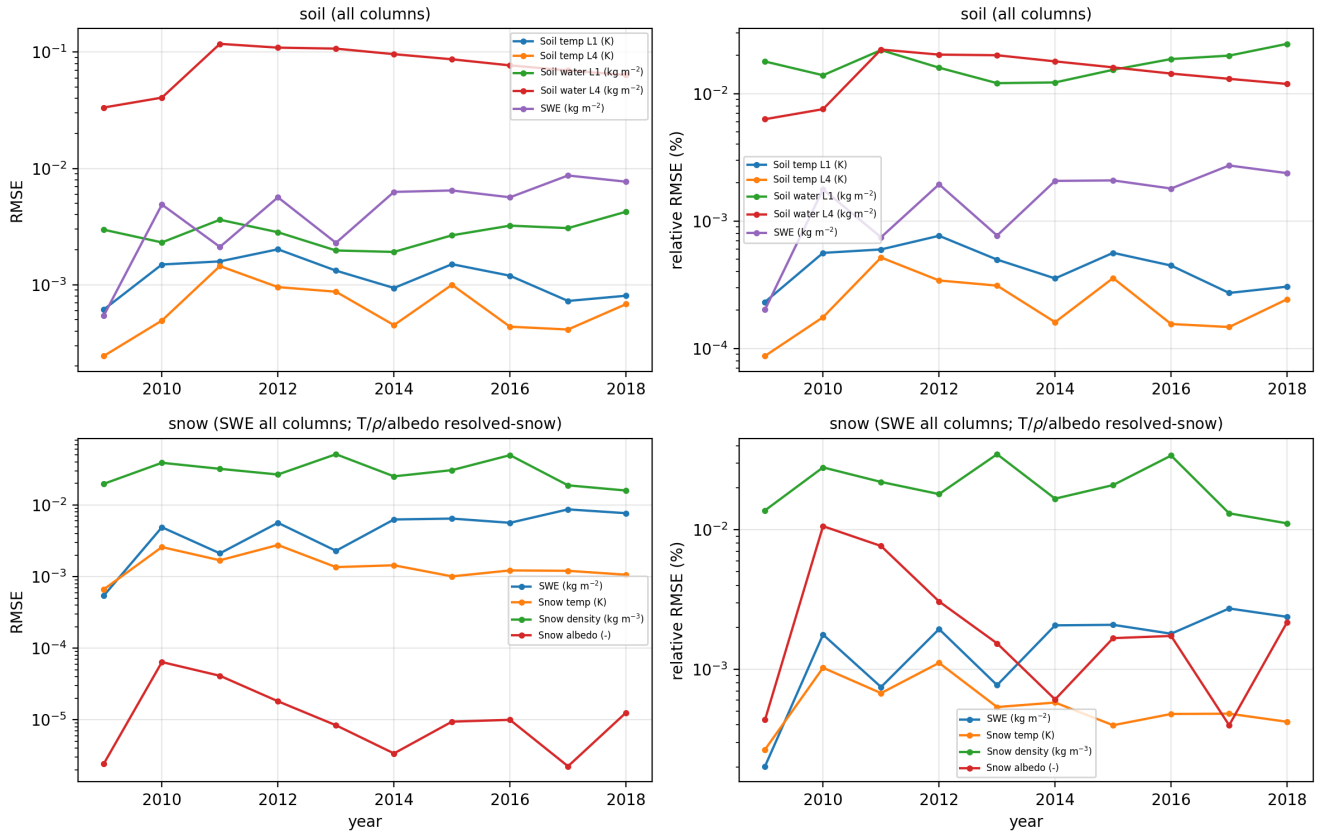
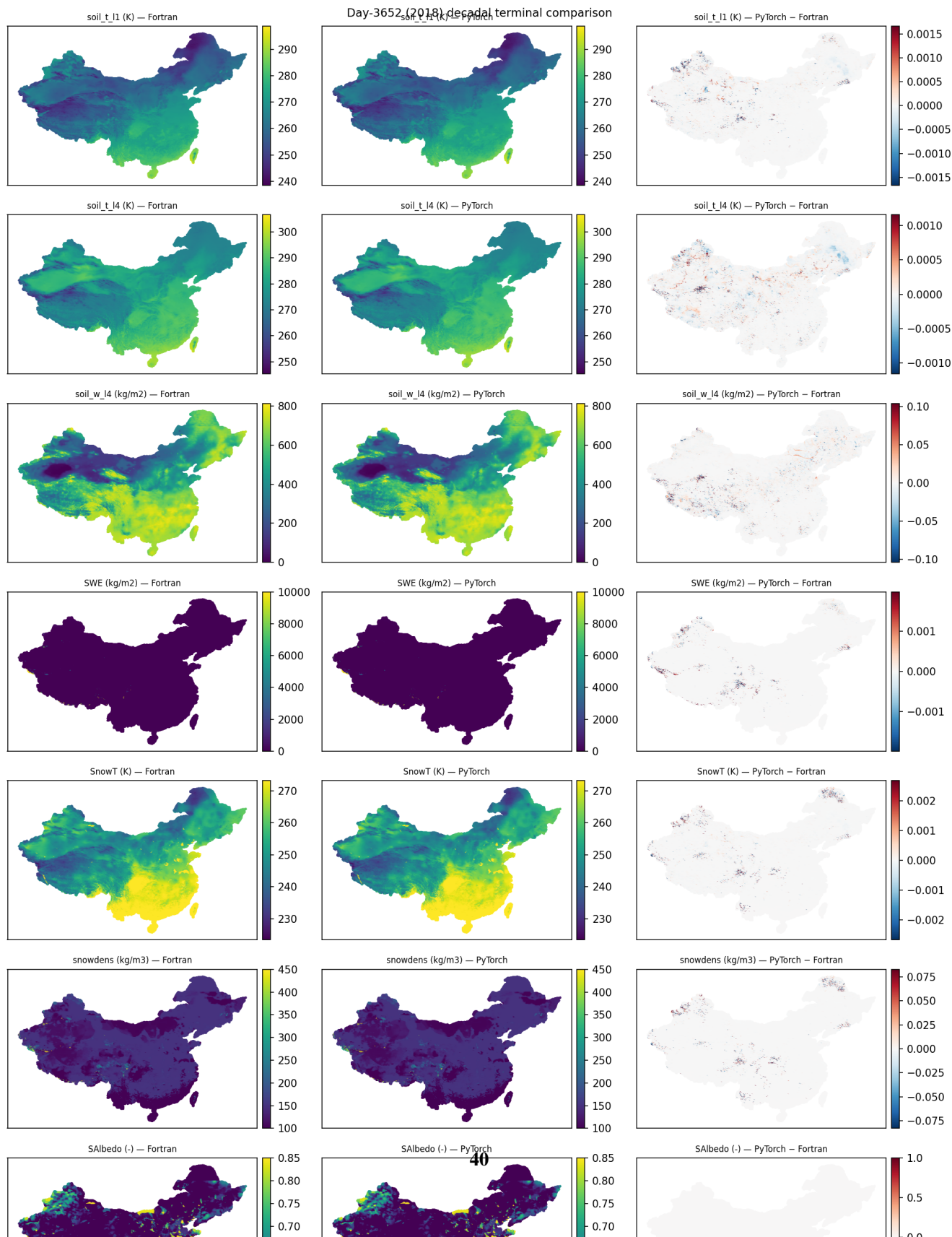


Figure A1. Fortran-PyTorch errors at the ten annual (31 December) terminal states of the successively restarted 2009–2018 integration, on logarithmic axes. Top row: soil temperature (layers 1–4) and soil water (layers 1–4) over all columns. Bottom row: snow water equivalent over all columns, and snow temperature, density, and albedo over the resolved-snow mask (snow-cover fraction above 10^{-3}). Left column: RMSE in native units; right column: relative RMSE.



Appendix B: Supporting differentiability experiments

B1 Finite-difference step dependence

Figure B1 shows the relative discrepancy between the reverse-mode derivatives of Table 3 and centred finite differences as a function of the perturbation magnitude. The broadly flat errors for the temperature and LAI controls and the shallow optimum for soil water are consistent with a resolved centred-difference approximation in double precision.

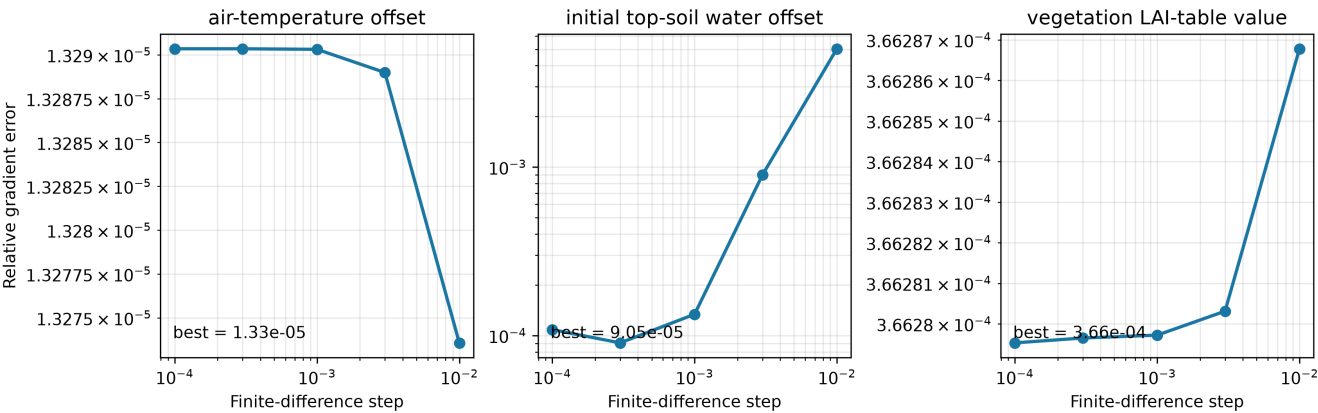


Figure B1. Relative discrepancy between reverse-mode automatic derivatives and centred finite differences for the 24 h SURF trajectory. Each panel uses the same loss and control as the corresponding row in Table 3.

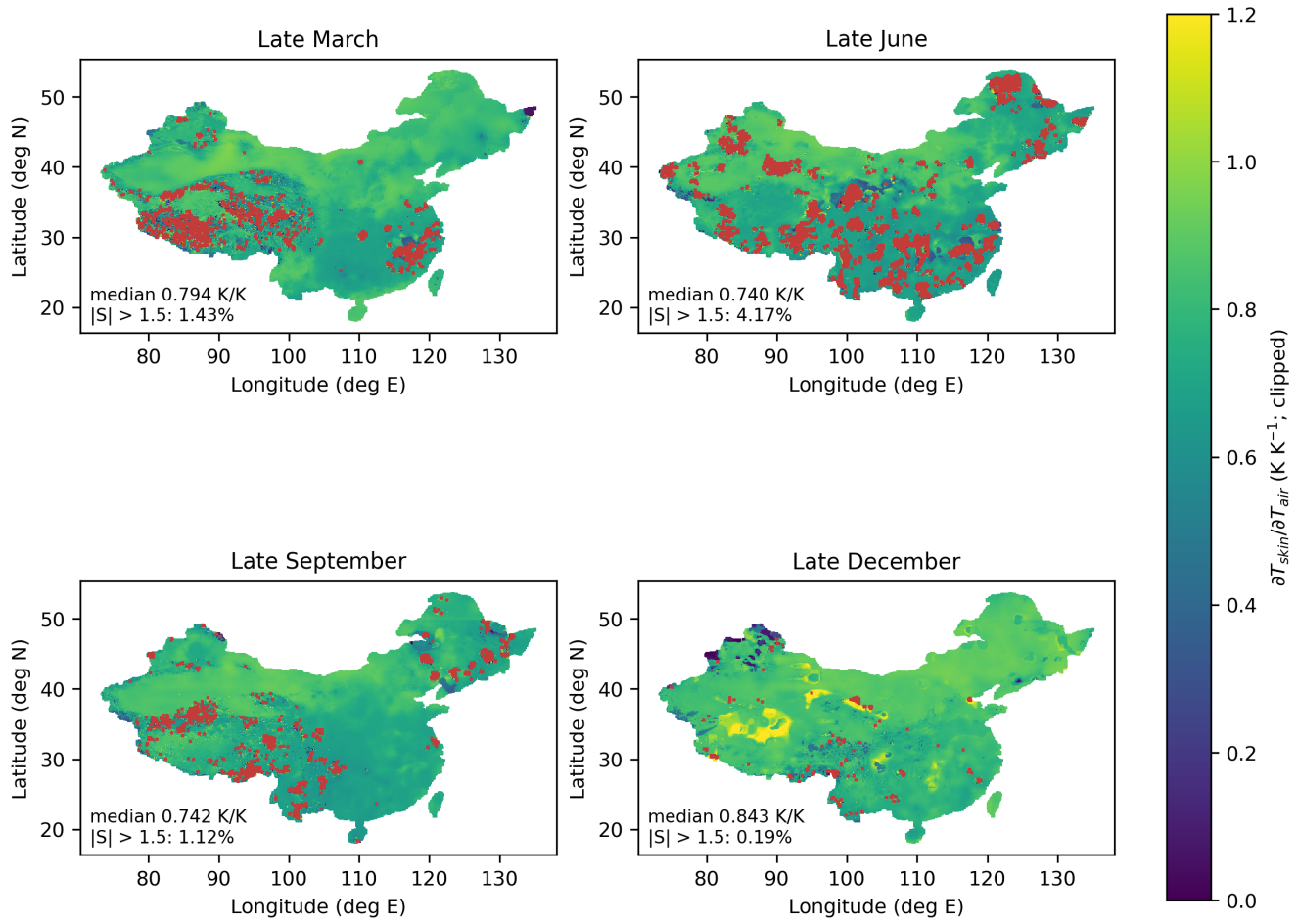


Figure B2. Seasonal local response of terminal skin temperature to an independent near-surface air-temperature offset, calculated by reverse-mode automatic differentiation over a 6 h CMFD window following a 24 h unperturbed closure adjustment. Colours show $\partial T_{\text{skin}} / \partial T_{\text{air}}$ between 0 and 1.2 K K^{-1} ; red pixels mark the separately reported threshold-sensitive columns with an un-clipped absolute response above 1.5 K K^{-1} .

B2 Fortran-guided initial-state retrieval

The verified derivative also admits an inverse calculation against an independent target. The target trajectory is the separately executed Fortran offline integration of the autonomous CMFD experiment, rather than a PyTorch synthetic twin. A representative warm, snow-free column at 23.05° N , 107.55° E was selected from the Fortran day-10 and day-20 states alone using a fixed median-state rule; the day-10 values of top- and second-layer soil water and top-layer soil temperature form the calibration target, and the day-20 state is held out for validation. Starting from the common restart with a known $+5.0 \text{ kg m}^{-2}$ perturbation of the initial top-layer soil water, the residual $r_i(d) = [h_i(d) - y_{\text{F},i}] / s_i$ (scales s_i of 5 and 10 kg m^{-2} and 1 K) is minimized

with a damped Gauss–Newton update whose Jacobian is obtained by reverse-mode automatic differentiation through the full 10-day trajectory; no finite-difference integrations enter the update.

Five updates reduced the imposed bias from 5.0 to 0.170 kg m^{−2} (Fig. B3); at day 10 the absolute errors of the three calibrated variables decreased by 93.9–95.9 %, and the held-out day-20 Fortran state independently confirms a 96.1–96.2 % reduction for all three variables (Table B1). The unperturbed common restart remains a numerical reference, with day-10 differences below 3.23 × 10^{−6} kg m^{−2} and 1.21 × 10^{−5} K, so the retrieval does not conceal a structural Fortran–PyTorch offset; it recovers a prescribed initial-condition error using a target external to the differentiated model. The experiment is a process-level demonstration rather than an observation-assimilation result; extending it will require observation operators, uncertainty models, and multi-column controls.

Table B1. Absolute PyTorch errors relative to the Fortran initial-state retrieval target. The day-10 values enter the Gauss–Newton objective; the day-20 values are held out. “Biased” is the prescribed +5.0 kg m^{−2} initial top-soil-water perturbation, and “retrieved” is the state after five updates.

Quantity (unit)	day-10 biased	day-10 retrieved	day-20 biased	day-20 retrieved
Soil water, layer 1 (kg m ^{−2})	4.04×10^{-1}	1.64×10^{-2}	3.01×10^{-1}	1.17×10^{-2}
Soil water, layer 2 (kg m ^{−2})	1.15	4.70×10^{-2}	6.88×10^{-1}	2.64×10^{-2}
Soil temperature, layer 1 (K)	2.68×10^{-3}	1.62×10^{-4}	1.28×10^{-2}	4.89×10^{-4}

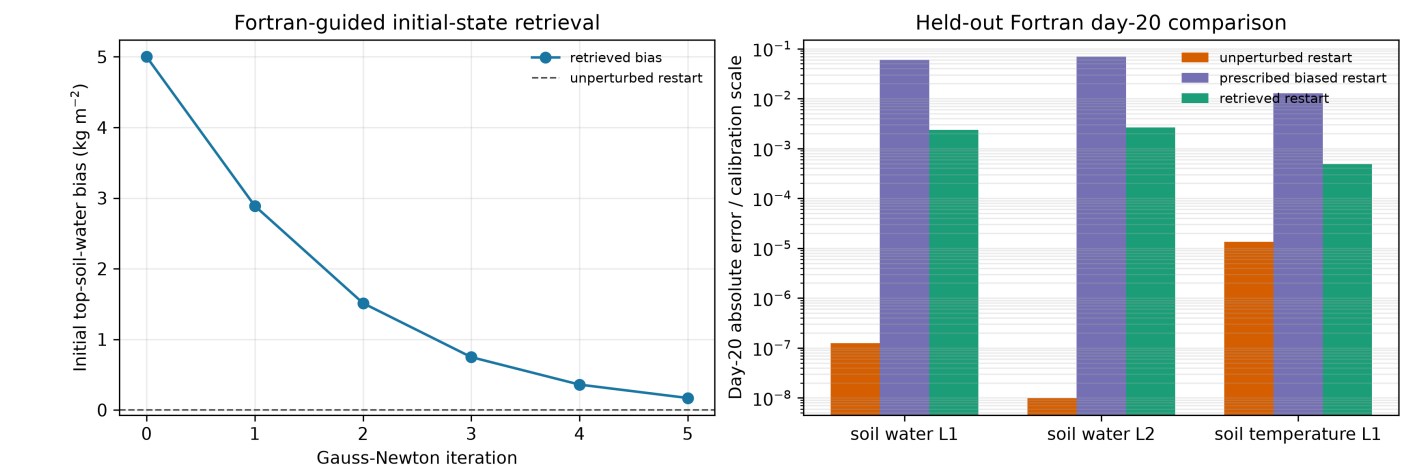


Figure B3. Fortran-guided initial-state retrieval using an autonomous offline SURF trajectory. Left: reverse-mode, damped Gauss–Newton recovery of the prescribed top-layer soil-water bias from the day-10 Fortran target. Right: normalized absolute errors against the held-out day-20 Fortran state. The unperturbed restart is shown only as a numerical-reference path and is not used to construct the retrieval target.

Appendix C: Parameter calibration and pseudo-forcing attribution against MODIS LST

This appendix documents the gradient-based experiments that precede the surface-layer closure of Sect. 5.2: an observation-space information analysis, an observing-system simulation, a parameter calibration against real MODIS observations, a pseudo-forcing attribution, and the forward timescale-asymmetry sweep that brackets the closure’s two relaxation times. Their role is diagnostic rather than corrective: the table-scale calibration removes little of the bias in either window, and the pseudo-forcing profile, though seasonally robust under the anchored evaluation, is an additive forcing adjustment with no process form. Together they establish the two facts the closure builds on: the reverse-mode inversion machinery is sound, and the daytime cold bias is structural, with a specific diurnal shape that no admissible parameter correction supplies; Sect. 5.1 states this attribution in the main text. All experiments use the spun-up initial state of Sect. 5, ten autumn–winter calibration days and ten held-out spring–summer days, the strict-QC day and night MODIS channels, and $\cos\phi$ area weighting; the closure of Sect. 5.2 reuses the same protocol.

C1 Observation-space parameter identifiability

The existence of a reverse-mode derivative does not imply that a parameter can be inferred from a given observation set. We therefore used the MODIS LST operator of Sect. 5 to diagnose the local information content before attempting parameter optimization. Four seasonally matched one-day windows were initialized from the corresponding annual state archive, each after a 24 h unperturbed closure adjustment, with the model LST sampled at the cell-specific MODIS view times under the strict QC mask and $\cos\phi$ area weights. The control vector comprised three dimensionless, logarithmic scaling directions applied to the vegetation lookup tables (leaf-area index, minimum stomatal resistance, momentum roughness length). If $\mathbf{r}$ is the area-weighted, QC-filtered LST residual vector and $\mathbf{J} = \partial\mathbf{r}/\partial\log\mathbf{s}$, the local Gauss–Newton information matrix is $\mathbf{F} = \mathbf{J}^\top\mathbf{J}$, estimated with 16 fixed-seed Rademacher reverse-mode projections.

The information content was evaluated separately for the spring–summer and the autumn–winter windows. The spring–summer information eigenvalues were 1.57×10^7 , 4.41×10^2 , and 53.0, giving a condition number of 2.96×10^5 ; the autumn–winter eigenvalues were 54.8, 5.65, and 0.823, with a condition number of 67. The strong spring–summer collinearity is also visible in Fig. C1: the normalized leaf-area–stomatal-resistance information correlation is -0.99 , whereas the autumn–winter correlations are much smaller in magnitude. The same differentiable model is thus locally ill-conditioned in one season and substantially better constrained in another, a property of the physics and observing pattern, not of the optimizer, and it dictates the design of the calibration.

C2 Conditioned-window calibration against MODIS LST

Guided by the identifiability diagnostic, the seasonal split is inverted: multiplicative parameter-table scales are fitted in ten well-conditioned autumn–winter windows and evaluated on ten held-out spring–summer windows that never enter the loss (2018; days of year 271, 281, ..., 361 and 91, 101, ..., 181, respectively, the same window set as the closure calibration of Sect. 5.2). Each window restores the land state from the daily archive of the autonomous, unclosed 2018 integration at the

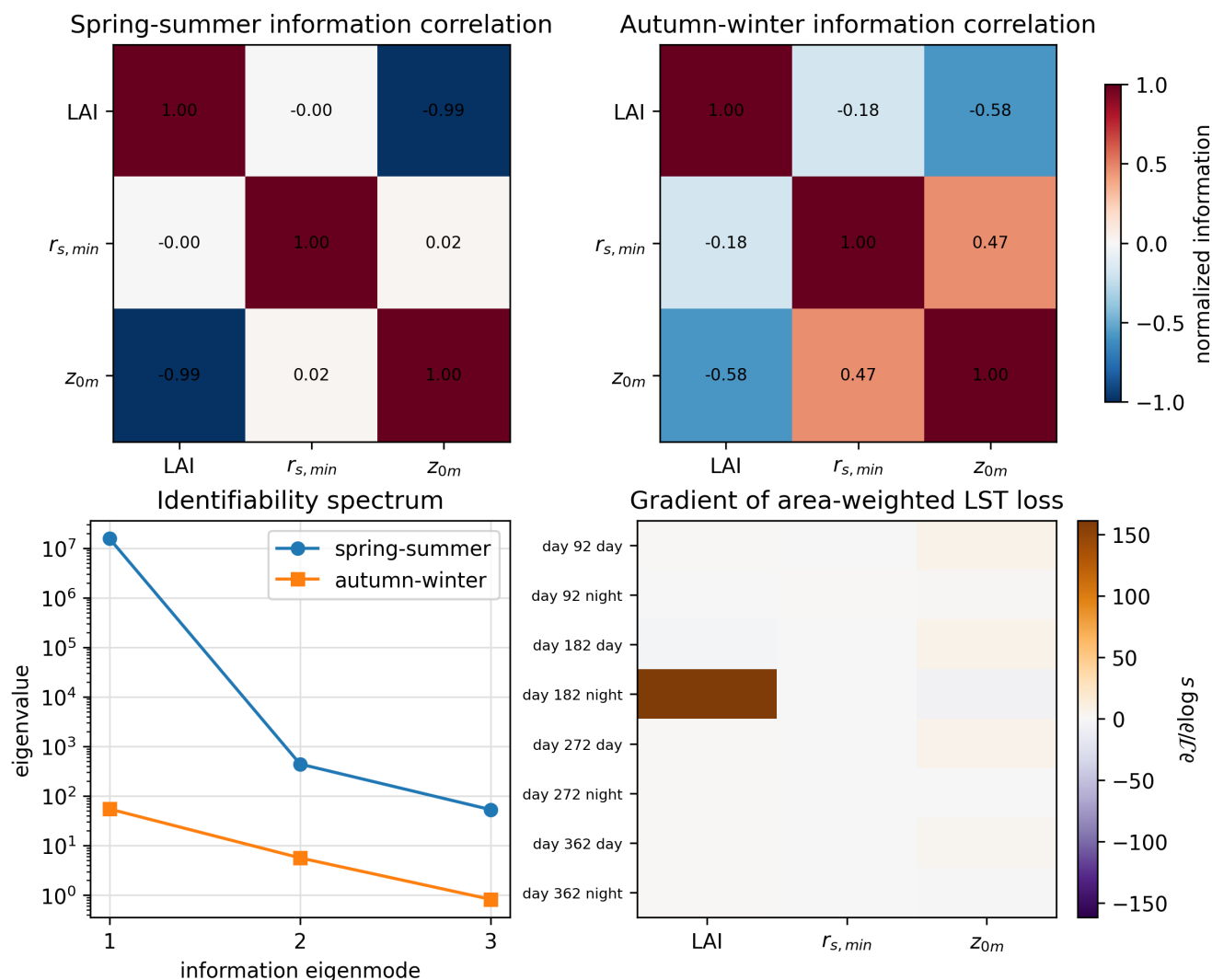


Figure C1. MODIS LST observation-space identifiability of three logarithmic vegetation scaling controls. Upper panels show normalized information correlations for the spring–summer and the autumn–winter windows. Lower left gives the corresponding information eigenvalues; lower right gives the seasonal LST-loss gradients. The matrices are area-weighted and computed from strict MODIS QC samples.

750 previous day boundary, advances 48 physics steps without gradients so that fast variables re-equilibrate, and then integrates a further 48 steps with the radiative skin temperature sampled at each column’s strict-QC MODIS overpass time, separately for the day and night passes. Training uses every fifth CMFD column (19,542 of 97,709; column-independent offline dynamics are unchanged), and every reported verification statistic uses the full domain: 73,700 daytime and 15,491 nighttime calibration samples, and 131,030 and 62,077 held-out samples.

755 The control vector contains eight multiplicative table scales: the leaf-area-index, minimum-stomatal-resistance, and roughness-length vegetation tables, each split between the domain’s high- and low-vegetation classes, a joint scale on the dry and saturated soil thermal-conductivity end-members, and a scale on the dry-soil heat-capacity table. Writing s_j for the logarithm of scale j , the affected table entries are set to $\bar{T}_{j,e} e^{s_j}$ on the class mask, and the loss aggregates the area-weighted mean-squared LST residual over all strict-QC samples of the ten calibration windows,

$$760 \quad \mathcal{J}(\mathbf{s}) = \sum_{d \in \mathcal{C}} \sum_{p \in \{\text{day}, \text{night}\}} \frac{\sum_i \cos \phi_i [T_i^{\text{mod}}(t_i; \mathbf{s}) - T_i^{\text{obs}}]^2}{\sum_i \cos \phi_i}. \quad (\text{C1})$$

One reverse pass through the 48 gradient steps differentiates the loss with respect to every table element, from which the scale derivatives follow by the chain rule, so a single forward-backward calculation per window trains all eight controls simultaneously. The scales are updated with Adam (learning rate 0.02) and bounded in $[1/2, 2]$, and the state is freshly restored from the archive every iteration, so no parameter-induced drift accumulates.

765 A no-gradient leverage probe on the first calibration day fixes the control selection: the three high-vegetation scales change the loss by exactly zero on this snow-free day (the structural zero-gradient case of Sect. 4.3), while the low-vegetation roughness scale dominates ($|\Delta \mathcal{J}| \approx 0.97$ per $\pm 10\%$, an order of magnitude more than any other active control), which motivates splitting the vegetation tables by class. In an observing-system simulation with the real operator, overpass times, and QC masks, synthetic observations from known truths (roughness scale 0.70, soil heat-capacity scale 1.40) plus 1 K noise, the documented
770 MODIS accuracy, the same loop recovers the truths to within 0.3–1.1 % (roughness recovered to 0.692, heat capacity to 1.404), drives the training loss to its analytic noise floor, and transfers to the held-out days; the two unperturbed but collinear controls drift to compensating values, exactly the collinearity diagnosed above (Fig. C2). The inversion machinery is therefore sound, and any remaining misfit against real observations is not an optimizer artefact.

Against real observations the same loop converges, but the achievable reduction is bounded by structural error rather than
775 by the optimizer (Table C1 and Fig. C3). The fitted scales lean coherently against the daytime cold offset: both momentum-roughness scales fall to about half their prior (0.56 in the high- and low-vegetation classes alike), weakening daytime turbulent exchange; both minimum stomatal resistances rise (1.50 and 1.62), suppressing transpiration; both leaf-area scales fall (0.69 and 0.66); and the soil-conductivity scale rises to 1.88, near its bound, amplifying the ground heat flux, while the soil heat-capacity scale falls to 0.71. The day-pass RMSE improves by 5.7 % in the calibration window and transfers at 3.8 % held-out,
780 with the night passes essentially unchanged (-2.0% and -0.5%). Repeating the fit from the spun-up initial state with doubled budget and a widened $[1/4, 4]$ admissible box removes no additional bias (the roughness scales settle at an interior optimum of 0.46–0.48): of the 5.5–8.3 K daytime cold offset, these eight global multiplicative table scales remove only about 0.5 K (7–10 %), in either window, under either box. This bounds the leverage of *these global scalings on land surface temperature*; it does not exclude a larger role for other parameters or for other state variables. The scales can only lean against the offset; the
785 remainder is associated with the process formulation.

Table C1. Full-domain MODIS LST verification over all 97,709 columns for prior and calibrated table scales (extended control vector). The calibration window aggregates ten autumn–winter days; the held-out window aggregates ten spring–summer days that never enter the loss.

Window	Pass	Prior		Calibrated (extended)	
		bias (K)	RMSE (K)	bias (K)	RMSE (K)
Calibration (autumn–winter)	day	−5.482	7.446	−4.955	7.022
	night	−0.428	3.834	−0.554	3.758
Held-out (spring–summer)	day	−8.282	11.342	−7.733	10.912
	night	+1.072	4.349	−0.113	4.326

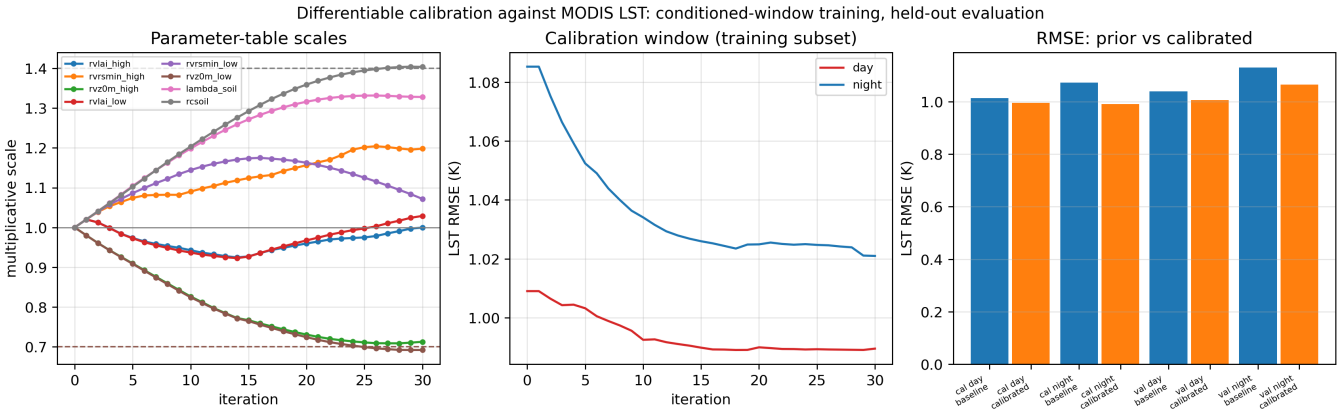


Figure C2. Observing-system simulation experiment with the real MODIS operator, overpass times, and QC masks (1 K observation noise on the training subset). Left: Adam trajectories of the eight table scales; dashed lines mark the imposed truth values, and the two collinear unperturbed controls visibly drift to compensating values. Centre: LST RMSE over the training subset, converging to the noise floor. Right: prior-versus-calibrated RMSE in the calibration and held-out windows.

C3 Pseudo-forcing attribution

The control vector of the final experiment is pseudo-forcing: eight additive air-temperature offsets, one per local-time 3-hour bin, plus multiplicative wind and shortwave scales (ten controls). A one-day finite-difference probe already locates the leverage: the Terra day-overpass bin (09:00–12:00 local) and the shortwave scale dominate ($|\Delta\mathcal{L}| \approx 4.1$ and 7.1 per $+0.5$ K and $+10\%$), the two nocturnal bins respond with opposite sign (warming them degrades the night pass), and the early-morning bins are nearly inert. Over 40 Adam iterations the calibration-window day bias is largely removed (-5.48 to -1.44 K; RMSE 7.45 to 5.11 K) at the optimal profile afternoon $+4.8$ K, evening -3.1 K, with the shortwave scale unmoved at 1.0 and the wind scale settling at 0.72 : the radiative correction is redundant once the air-temperature profile is free to move. The profile states what the offline configuration lacks: a near-surface air-temperature diurnal amplitude about 8.1 K peak-to-peak larger than the reanalysis delivers, or an equivalent radiative gain. Evaluated in the held-out spring–summer window under the same anchored-day protocol, the corrections retain most of their effect (day bias -8.28 to -3.97 K; day RMSE 11.34 to 8.59 K, night RMSE

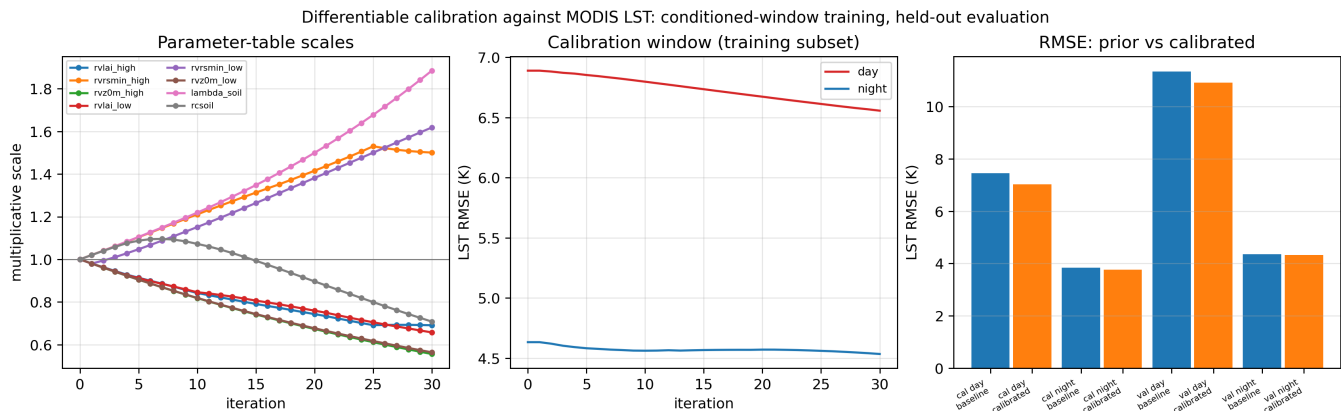


Figure C3. Differentiable calibration against real MODIS Terra LST with an identifiability-guided seasonal split (extended control vector). Left: trajectories of the eight table scales; the roughness scales migrate toward the lower admissible bound while compensating the structural daytime cold bias. Centre: area-weighted LST RMSE over the training subset in the autumn–winter calibration window. Right: full-domain prior-versus-calibrated RMSE in the calibration and the held-out spring–summer windows, day and night passes.

4.35 to 3.66 K), so the diagnosed profile is seasonally robust (Fig. 11). It is not, however, a parameterization: it is a ten-control additive adjustment of one year’s forcing fitted to one sensor, with no process form that could enter the model physics or transfer to another forcing set. The diagnostic value of the experiment lies in the shape of the required correction, not in the number.

C4 Per-channel forcing sensitivity of daytime LST

The pseudo-forcing experiment of Sect. C3 assigns leverage to air temperature and shortwave among ten controls. A complementary single-channel check quantifies how strongly each forcing channel of Eq. (3) drives daytime land surface temperature on its own. For each channel in turn, a single physically meaningful perturbation is applied and the resulting change in domain-mean day-pass LST is measured over a strong-insolation summer day (day of year 181) from the common spun-up restart. Because the perturbations have different natural units and amplitudes, the comparison is made on a normalized basis: the LST response per one-percent relative change of the channel about its summer domain-mean value. Table C2 gives the result. The exchange air temperature dominates by an order of magnitude over every other channel, about five times the next strongest (downward longwave) and thirty times the shortwave, while humidity, precipitation, and surface pressure are weak. This ordering is why the closure of Sect. 5.2 acts on the exchange air temperature: it is the channel through which the diagnosed daytime deficit is transmitted most strongly, and the radiative part of the deficit is energetically equivalent to a correction of that same amplitude.

Table C2. Single-channel forcing sensitivity of domain-mean day-pass LST (day of year 181, spun-up restart, all 97,709 columns). For each forcing channel a single perturbation is applied and the change in domain-mean day LST is measured. The final column normalizes by the perturbation size as a percentage of the channel’s summer domain-mean value, making the channels comparable.

Channel	Perturbation	LST shift (K)	Sensitivity (K per 1 %)
Air temperature	+1 K	+0.293	+0.863
Downward longwave	+10 W m ⁻²	+0.457	+0.160
Surface pressure	×1.01	+0.054	+0.054
Wind speed	×1.10	+0.508	+0.051
Specific humidity	×1.10	+0.390	+0.039
Downward shortwave	×1.10	+0.268	+0.027
Precipitation	×2.0	−0.142	−0.001

C5 Timescale-asymmetry sweep and the state-space stability domain

Before the two relaxation times of Sect. 5.2 were trained, a forward-only grid sweep with $\lambda_s = 0.05$ held fixed mapped the response surface over $\tau_{day} \in \{3, 5, 10\}$ h and $\tau_{night} \in \{1, 2, 3, 5\}$ h under the anchored-day protocol. Every statistically stable asymmetric combination improves on the tied closure in both windows; the day-pass improvement grows monotonically as τ_{day} decreases, and the grid optimum ($\tau_{day} = 3$ h, $\tau_{night} = 2$ –3 h) sits on the edge of the τ_{day} range, which motivated the gradient calibration of Sect. 5.2 over further grid refinement. The sweep also delivered the first stability information: $\tau_{night} = 1$ h is stable throughout the autumn–winter calibration window yet destabilizes every held-out spring–summer integration (day-pass biases of 10^5 – 10^8 K). Statistical screening alone, however, is not a sufficient acceptance test, and the unconstrained training endpoint (2.47, 2.36) h exposed why: it verifies on the calibration window and on every held-out day pass, yet its held-out integration of 11 May (day 131) diverges. We therefore re-verified the grid in state space with a step-level diagnostic that repeats the anchored-day protocol and records the anomaly ΔT itself for every physics step and every column, rather than view-time samples under quality control. The diagnostic is bitwise-reproducible across two GPU architectures (A100 and RTX A6000), so what follows is deterministic model dynamics, not hardware numerics. On day 131 the failure is localized (Table C3): 18 columns in two Tibetan-Plateau clusters (30–31°N, 84–86°E and 37–38°N, 87–88°E) carry an intrinsic two-step (~ 1 h) limit cycle in the grid-mean sensible heat flux H (amplitude about -100 to $+150$ W m⁻² at vanishing anomaly): the offline exchange flips between stability regimes through the dawn transition. The closure couples into this cycle through its quasi-steady target $-\lambda_s H$. For the unconstrained endpoint, the fatal columns respond at the dawn transition of the forecast day (steps 48–51, local 05:42–07:13), where the flux swings between about -120 and $+470$ W m⁻² over consecutive steps; the anomaly passes 6 K by step 55 and 10^4 K by step 67, and the first non-finite state appears at step 56, after which growth is superexponential. The domain is a resonance structure, not a monotone boundary: ($\tau_{day} = 4$ h, $\tau_{night} = 3$ h) diverges on day 131 while ($\tau_{day} = 5$ h, $\tau_{night} = 3$ h) peaks at 54 K and ($\tau_{day} = 4$ h, $\tau_{night} = 4$ h) at 2.5×10^2 K, so no axis-aligned interpolation of the statistical sweep could have located it.

Table C3. State-space verification of the closure grid on held-out day 131: maximum $|\Delta T|$ (K) over all columns and all 96 anchored steps. “Diverged” marks non-finite states. The combinations adopted for training also pass the same test on all 20 anchored evaluation days (worst single-column transient 109 K at $(\tau_{\text{day}}, \tau_{\text{night}}) = (6, 3)$ h, decaying); $\tau_{\text{night}} = 2$ h exceeds 10^3 K on day 341 at $\tau_{\text{day}} \in \{5, 8, 10\}$ h and on day 121 at $\tau_{\text{day}} = 5$ h. The tied closure (10.1 h, 10.1 h) peaks at 34 K on this day.

	$\tau_{\text{day}} = 3$ h	$\tau_{\text{day}} = 4$ h	$\tau_{\text{day}} = 5$ h	$\tau_{\text{day}} = 6$ h	$\tau_{\text{day}} = 8$ h	$\tau_{\text{day}} = 10$ h
$\tau_{\text{night}} = 2$ h	9.4×10^{20}	3.7×10^2	1.6×10^2	65	61	48
$\tau_{\text{night}} = 3$ h	9.8×10^7	diverged	54	78	51	29
$\tau_{\text{night}} = 4$ h	diverged	2.5×10^2	48	60	42	39
$\tau_{\text{night}} = 5$ h	diverged	2.1×10^2	64	65	44	29

835 The most consequential property of this failure is that the observation-space statistics cannot see it. The window aggregates of the sweep remain clean at divergent grid points, $(3, 2)$, whose day-131 state-space maximum is 9.4×10^{20} K, and $(10, 2)$, which exceeds 10^3 K on day 341, because the strict MODIS quality control removes exactly the divergent columns from the sampled statistics. A verification protocol that scores only quality-controlled observations at view times can therefore accept a configuration whose state space diverges; monitoring the state itself is what separates the two. Applying the step-level test to all
840 20 anchored evaluation days maps the domain: $\tau_{\text{night}} = 2$ h exceeds 10^3 K on at least one anchored day at $\tau_{\text{day}} \in \{5, 8, 10\}$ h (days 121 and 341, a few columns each) and stays finite but elevated at $\tau_{\text{day}} = 6$ h (worst transient 150 K); $\tau_{\text{day}} = 5$ h diverges on day 121 at $\tau_{\text{night}} \in \{2, 4, 5\}$ h while $(5, 3)$ stays finite (worst 259 K); and $(6, 5)$ diverges on day 121. The combinations $(6, 3)$, $(6, 4)$, and $\tau_{\text{day}} \in \{8, 10\}$ h at $\tau_{\text{night}} \in \{3, 4, 5\}$ h pass all 20 days with no column beyond 141 K, and the tied closure’s worst transient is 49 K: the verified set is not rectangular. These verified combinations form the training domain of the constrained
845 gradient calibration in Sect. 5.2, whose optimum rests at their fastest corner.

Appendix D: Transfer of the surface-layer closure to the Fortran reference

The closure of Sect. 5.2 acts on physical, grid-computed quantities only, the surface sensible heat flux and the air temperature entering the surface exchange, and is therefore not tied to the differentiable implementation. This appendix documents its transfer to the Fortran offline driver of the reference code base and the bitwise invariance of the inactive closure and its transfer.

850 D1 Implementation in the offline driver

The recurrence is added to the driver level (`cp91s.F90`), outside the SURF physics itself, in four changes: (i) a persistent anomaly array per grid point; (ii) immediately before each call to the SURF physics, the lowest-level air temperature argument at the incoming time level is incremented by the anomaly, and the increment is removed again after the call; (iii) the anomaly is advanced with the identical discretization as Eq. (15), $\Delta T_{n+1} = \beta_n \Delta T_n + (1 - \beta_n)(-\lambda_s H_n)$, with the shortwave-blended per-
855 step relaxation factor β_n evaluated from the driver’s own surface downward shortwave argument, and H_n the grid-mean surface sensible heat flux of the step just completed (tile-fraction weighted, downward positive, the same quantity the torch model

differentiates through, exported from the vertical-diffusion interface of the reference physics); and (iv) environment-variable switches with compiled-in defaults $\lambda_s = 0$ and the tied timescale, so the unmodified driver behaviour is the compiled-in default. The constants are shared with the torch implementation ($\rho c_p = 1.2 \times 1005.7 \text{ J m}^{-3} \text{ K}^{-1}$, $S_{ref} = 50 \text{ W m}^{-2}$, $\Delta t = 1800 \text{ s}$), and
860 the update ordering matches: the anomaly applied at step $n+1$ is built from the flux and shortwave of step n .

D2 Bitwise invariance of the inactive closure

With $\lambda_s = 0$ the patch must be numerically invisible. A one-day integration of the patched executable (48 steps, 97 709 columns) was compared record by record against an archive produced by the unpatched executable from the same restart and forcing: all 24 output variables of the 3-hourly history file agree bitwise, and the run log confirms the closure inactive
865 ($\lambda_s = 0$ reported at initialization).

D3 Matched-protocol evaluation

All Fortran evaluations use year-long integrations started from the common spun-up restart of Sect. 5; the calibrated constants of the torch configuration ($\lambda_s = 0.05 \text{ K (W m}^{-2})^{-1}$, $\tau_{\text{day}} = 6 \text{ h}$, $\tau_{\text{night}} = 3 \text{ h}$) enter the driver through its environment switches without re-tuning. Because the driver history is 3-hourly, the Fortran runs and a re-sampled torch evaluation sample the nearest
870 3-hourly record at each MODIS view time; the observation operator is the driver’s grid-mean skin temperature (`AvgSurfT`), which is by construction the same tile-fraction-weighted mean as the torch *radiative skin temperature*. Under this matched protocol the year-long unpatched Fortran integration and the torch integration of the prior configuration agree to within 0.001 K in bias and RMSE in all four window-pass combinations (e.g. held-out day bias -8.068 versus -8.068 K , RMSE 11.217 versus 11.218 K), so the MODIS error statistics of the two implementations are indistinguishable at this protocol before the closure
875 is activated. What is verified to date is the numerical fidelity of the transfer (bitwise invariance of the inactive closure) and the indistinguishability of the prior statistics; full-year Fortran statistics of the calibrated configuration ($\lambda_s = 0.05$, $\tau_{\text{day}} = 6 \text{ h}$, $\tau_{\text{night}} = 3 \text{ h}$) are not yet available and are not anticipated here. All conclusions of Sect. 5.2 rest on the torch implementation.

Code and data availability. The common spun-up initial state, the observation subsets, the Fortran reference outputs, the archived evaluation products behind every reported number, and the per-experiment manifests are collected in the accompanying data package (Yang
880 et al., 2026) at the Science Data Bank, <https://doi.org/10.57760/sciencedb.0125b>. The source code of the PyTorch implementation and the experiment scripts are available from the corresponding author upon request and will be released publicly upon publication. The raw driving datasets are public and are given by reference: CMFD (<https://doi.org/10.6084/m9.figshare.c.4557599>), ERA5-Land (<https://doi.org/10.24381/cds.e2161bac>), and MODIS MOD11C1/MOD10C1 (<https://doi.org/10.5067/MODIS/MOD11C1.061>, <https://doi.org/10.5067/MODIS/MOD10C1.061>). The native OpenIFS source code remains subject to ECMWF distribution constraints.

885 *Author contributions.* J. Yang implemented the PyTorch translation, ran the verification and performance experiments, and drafted the manuscript outline.

Competing interests. The authors declare that they have no conflict of interest.

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